

TESSERA

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We'd like to

Monitor

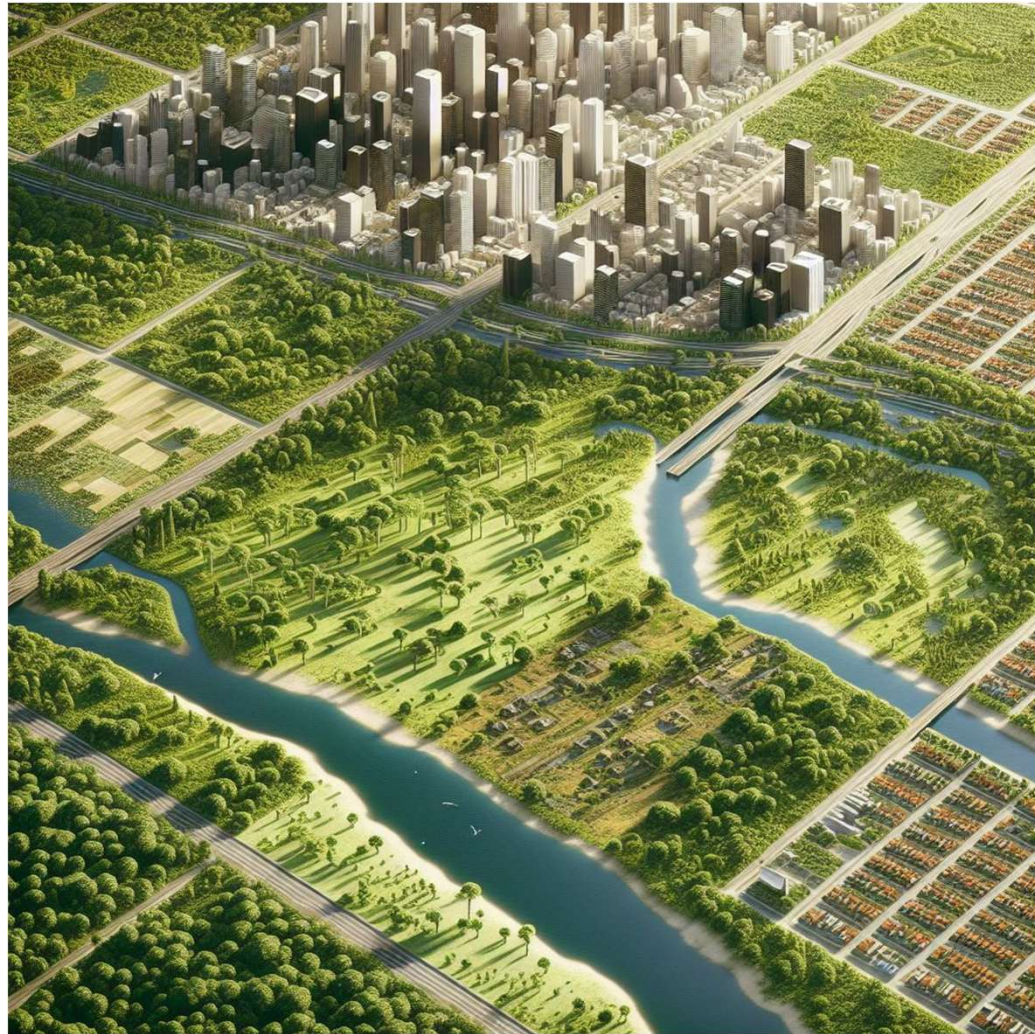
- Land use
- Biodiversity

Assess status of

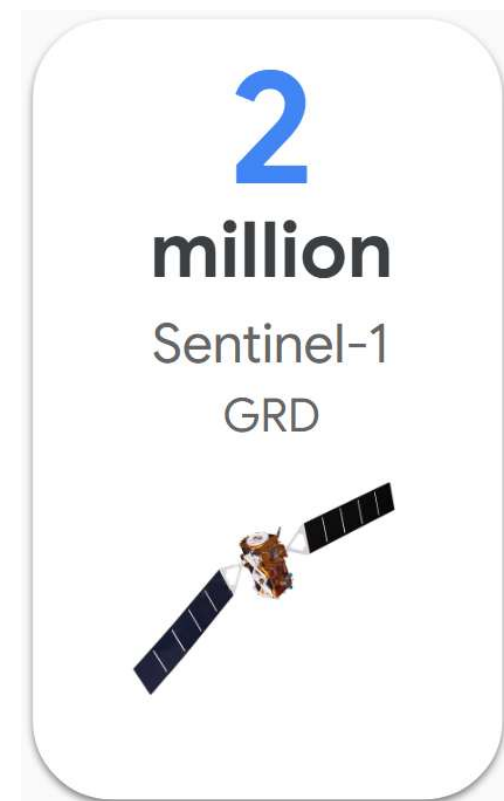
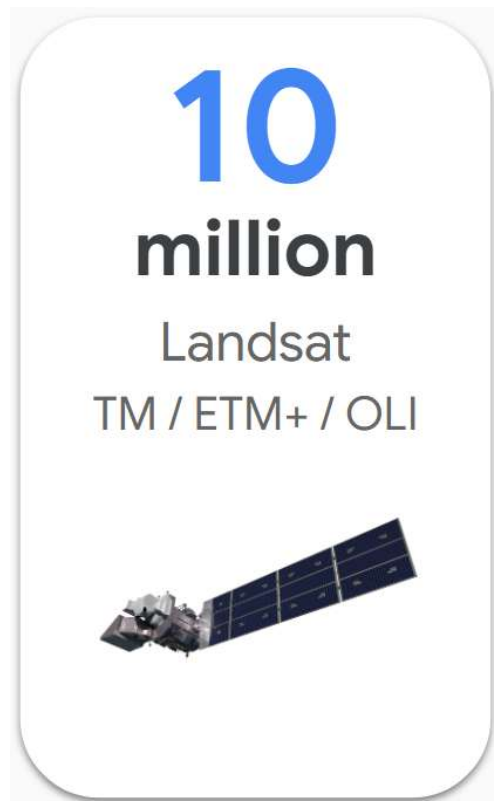
- Crops
- Soils
- Forests

Quantify

- Forest degradation
- Deforestation
- Carbon stock



Lots of public satellite data!



Problems

- Clouds
 - >50% in an image often cloud-covered; many models only accept <10%
 - Clouds block surface info, adding noise
- Varying time gaps
 - Irregular intervals hurt temporal learning
 - Large gaps = abrupt, hard-to-learn changes
- Changes in lighting
 - Lighting shifts mimic real land changes
 - Models may misclassify due to brightness



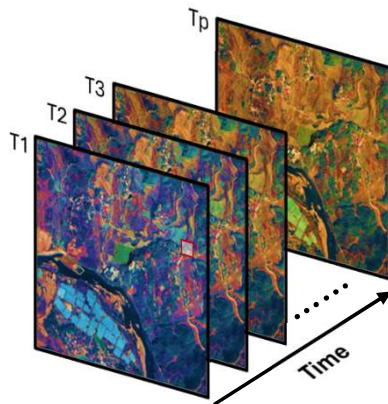
Current practice: **Compositing**

Multi-timestep
Cloud-corrupted

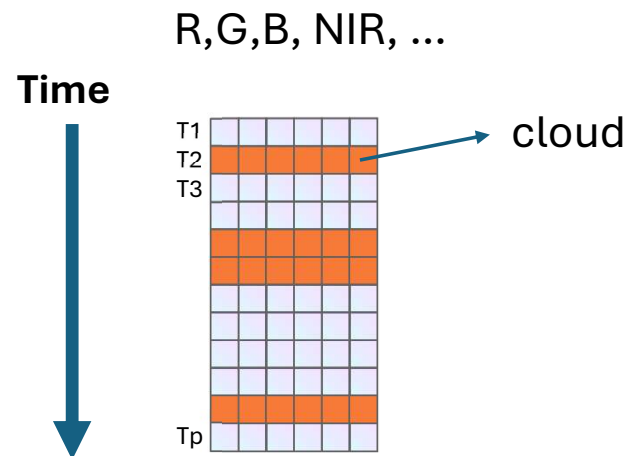


Single-/Few- timestep
Cloud-free

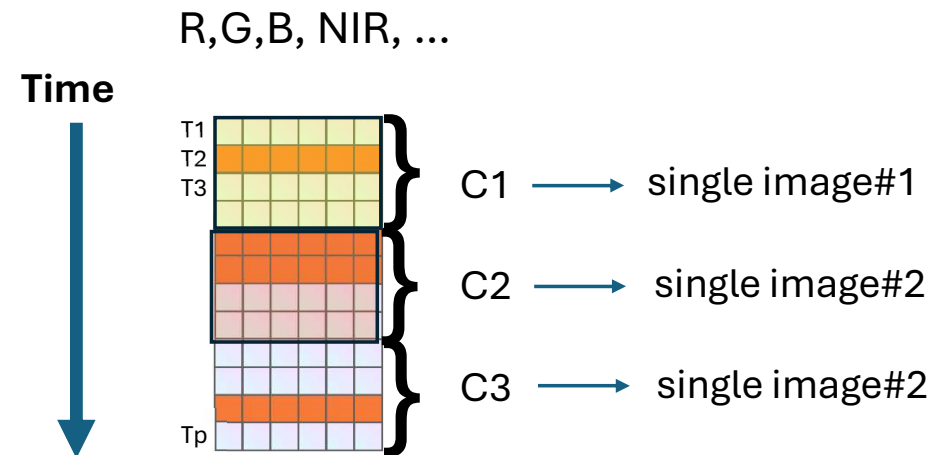
Cloud corrupted
Sentinel-2 MSI



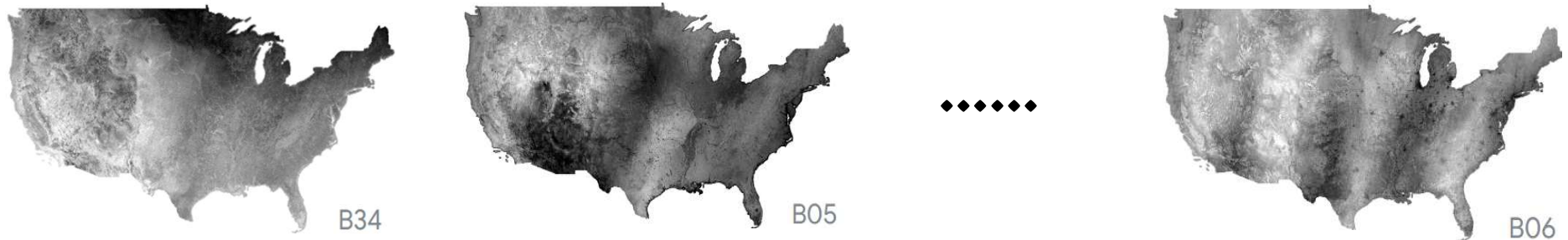
Problem with Compositing



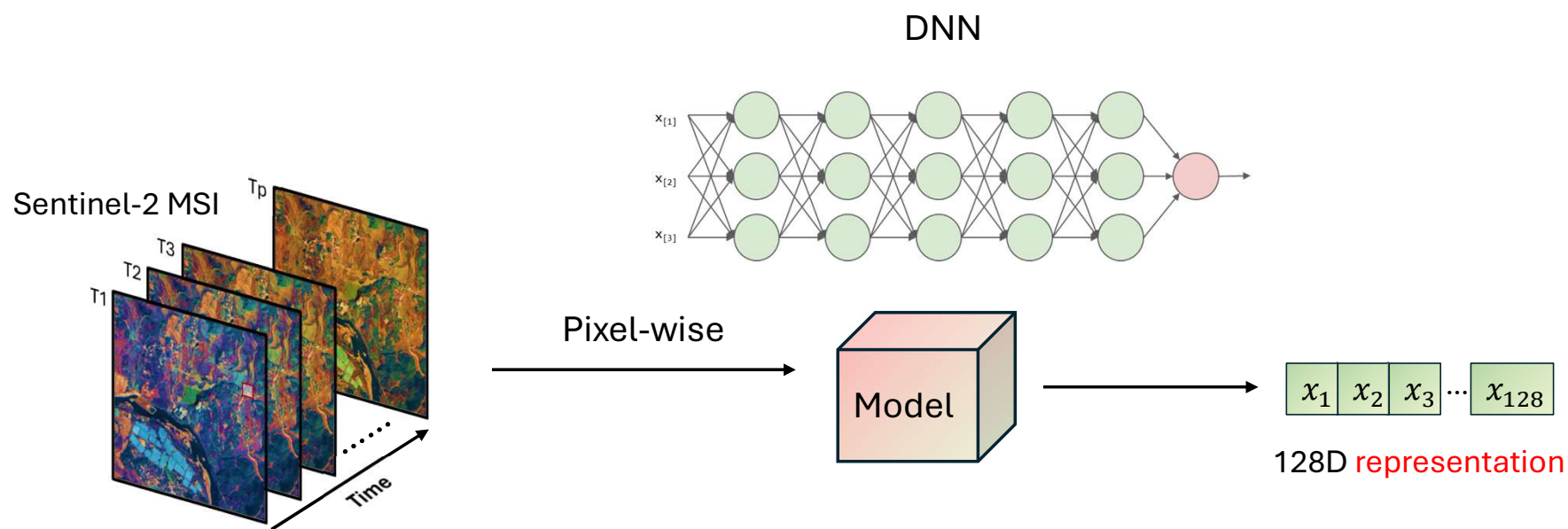
Compositing **hides** the **temporal signal**!



We use **AI** to produce a
128-dimension cloud-free “image”
encoding the temporal evolution of each spectral band



How?



Supervised Learning

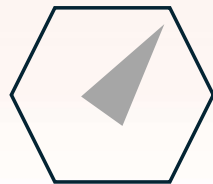
Data (x,y) where x is data, y is label

Goal Learn a function to map $x \rightarrow y$

Tasks

- Land Classification
- AGB Regression
- Object Detection

Performance



Self-supervised Learning (SSL)

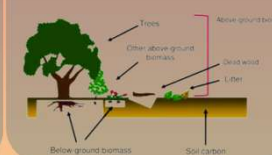
Only x!

Learn some underlying hidden structure of the data (**Representation**)

Land Classification



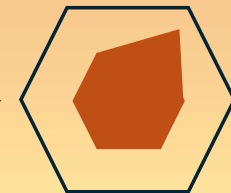
AGB Estimation



Object Detection

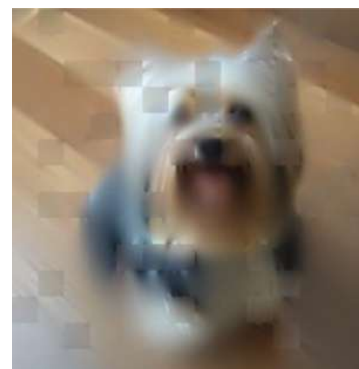
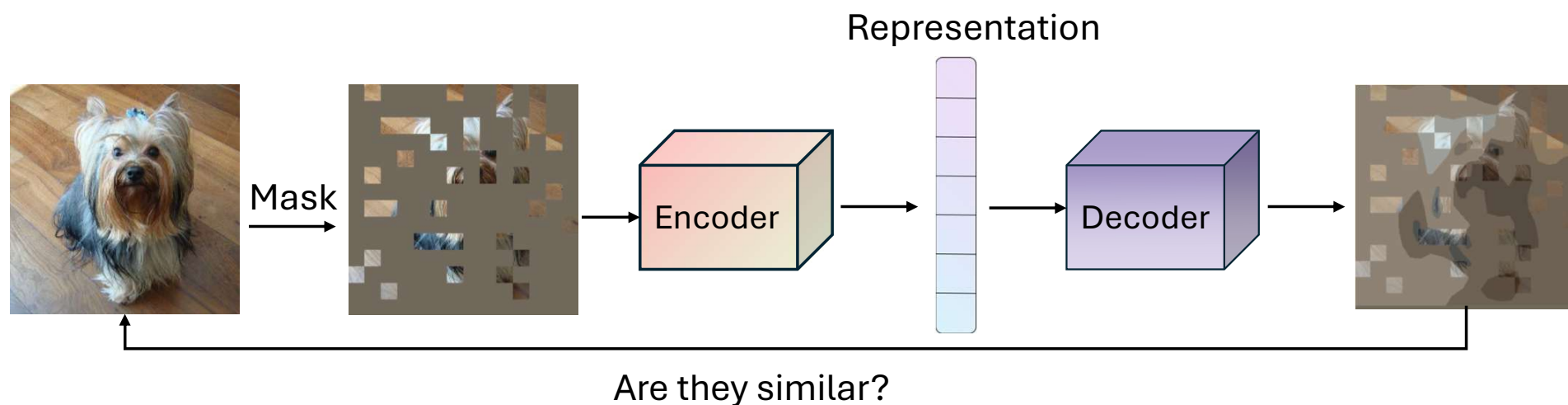


Finetune

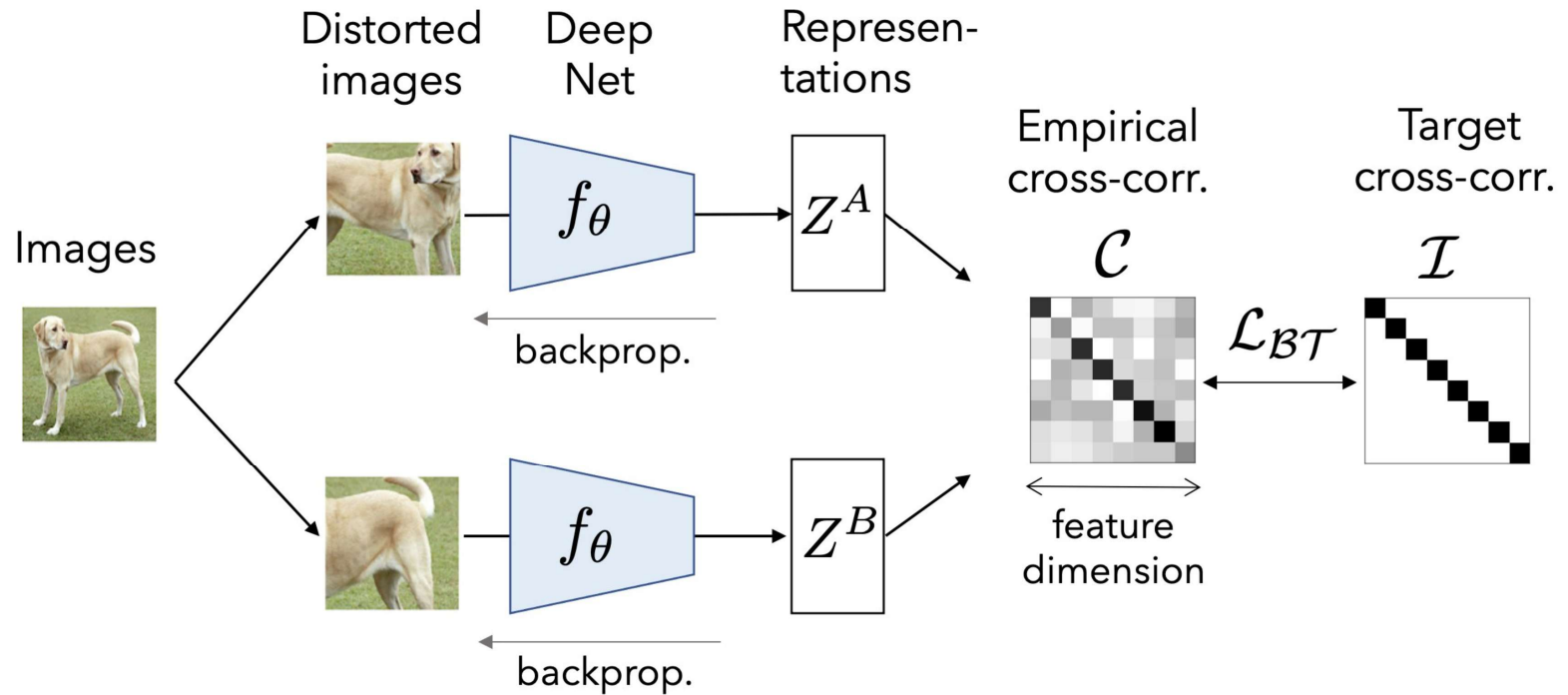


SSL example

Masked Auto Encoding (MAE)



Barlow Twins SSL*



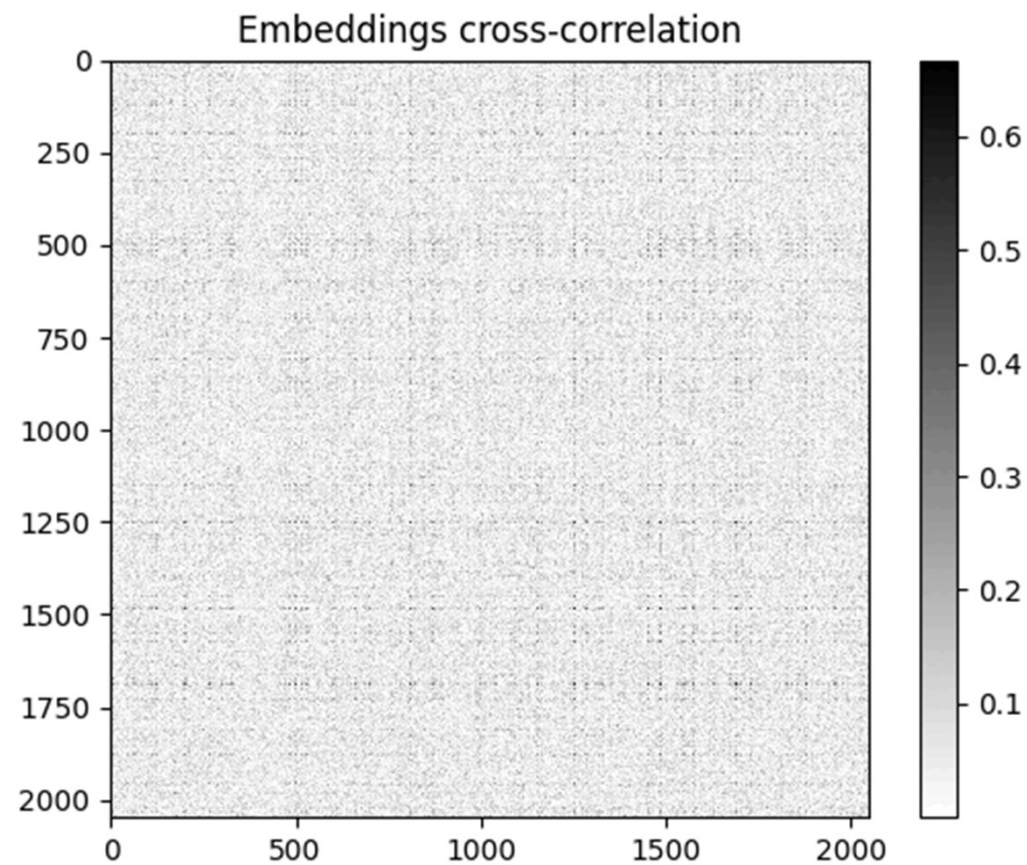
$$\mathcal{L}_{BT} \triangleq \underbrace{\sum_i (1 - C_{ii})^2}_{\text{invariance term}} + \lambda \underbrace{\sum_i \sum_{j \neq i} C_{ij}^2}_{\text{redundancy reduction term}}$$

$$C_{ij} \triangleq \frac{\sum_b z_{b,i}^A z_{b,j}^B}{\sqrt{\sum_b (z_{b,i}^A)^2} \sqrt{\sum_b (z_{b,j}^B)^2}}$$

* simplified

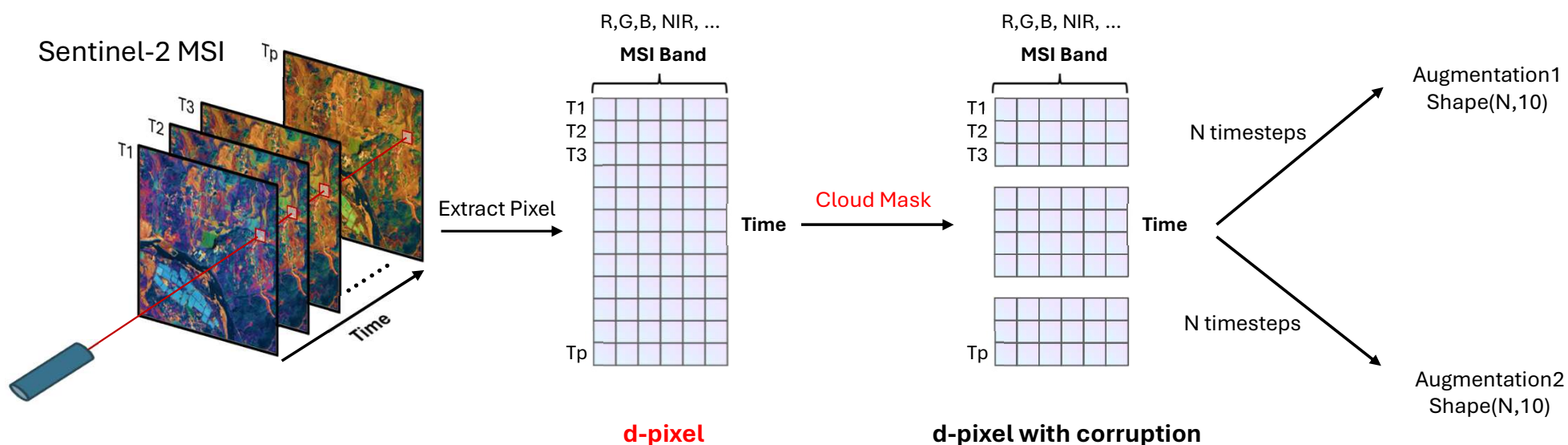


Cross-correlation Matrix



Barlow Twins for Remote Sensing

Create **two** augmentations (sub-samples) from the **same** sample

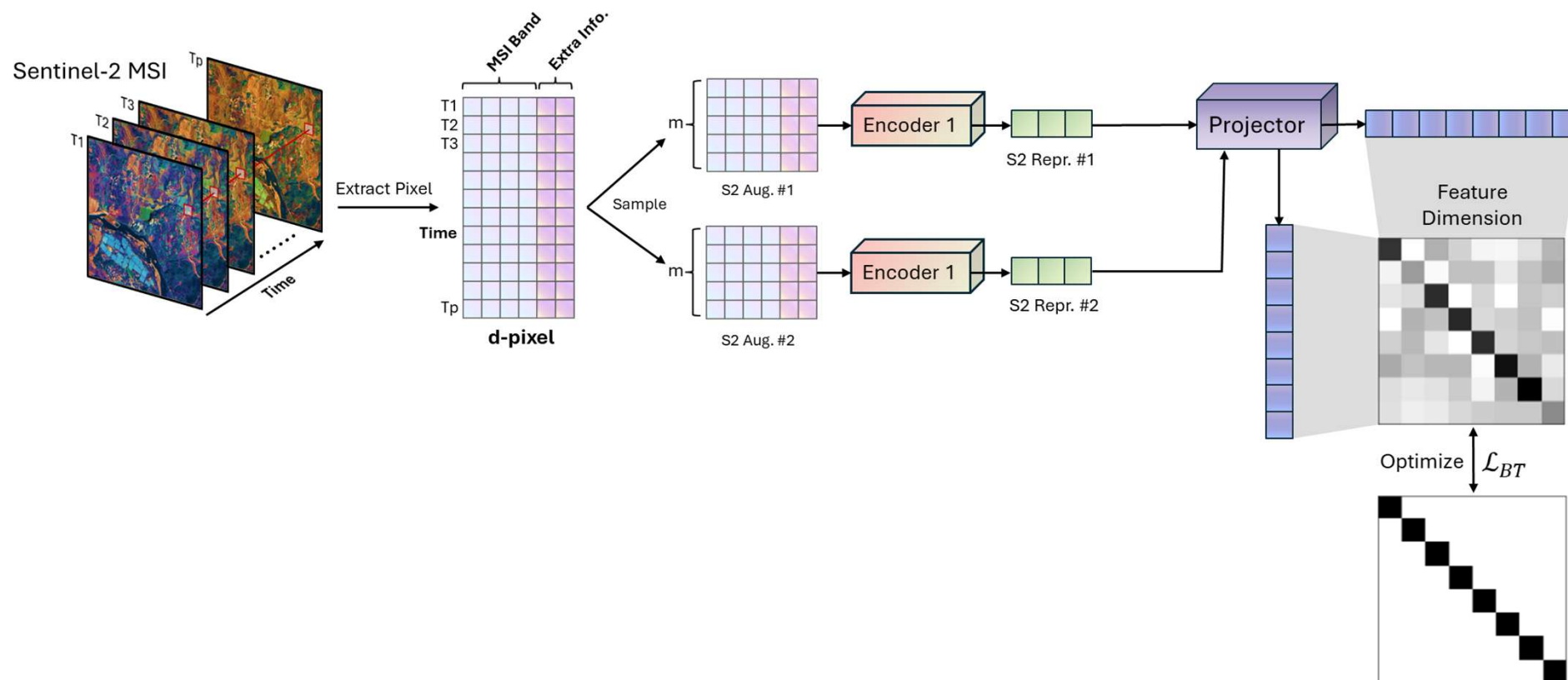


1.Introduction

2.Methodology

3.Result

4. Future Work



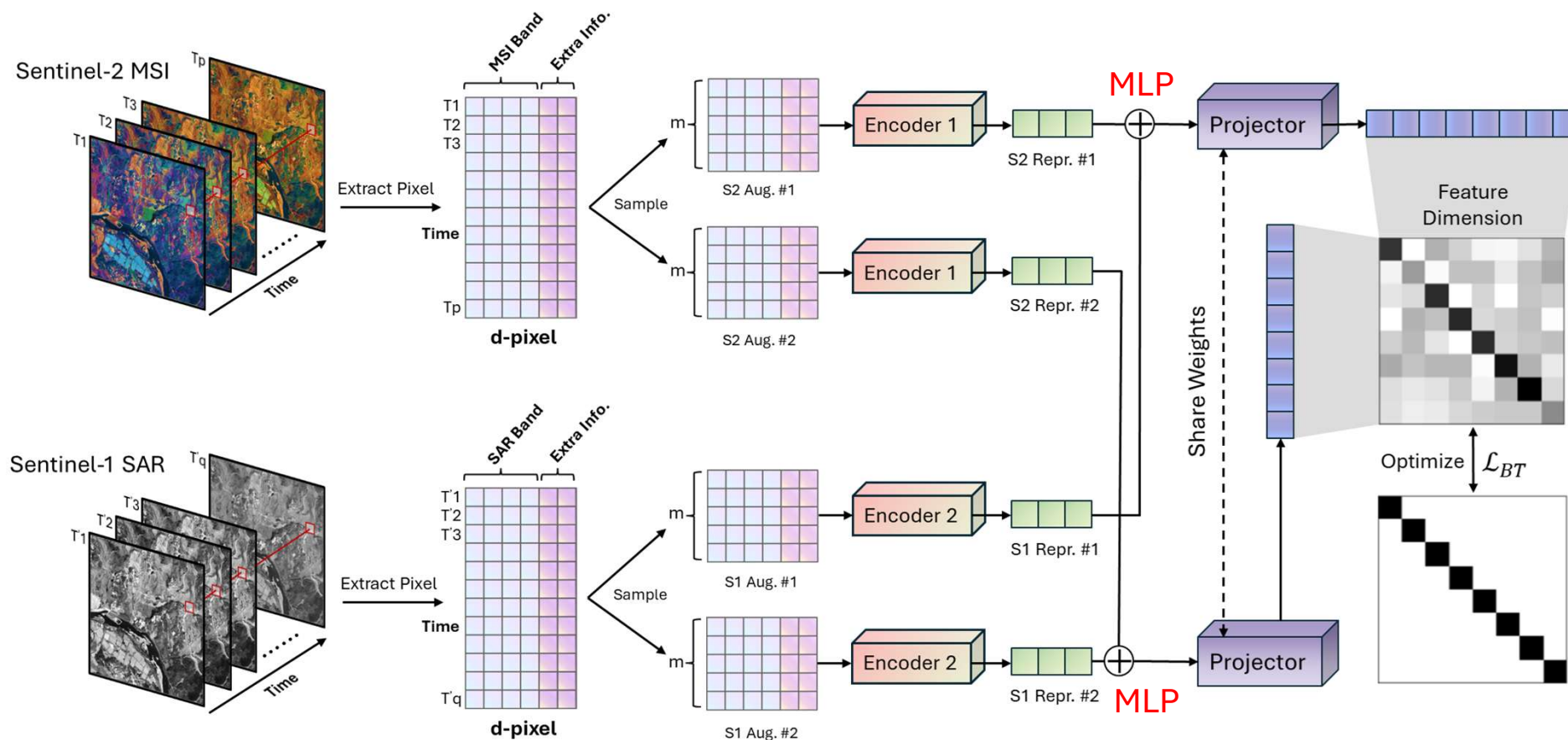
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Add Sentinel 1

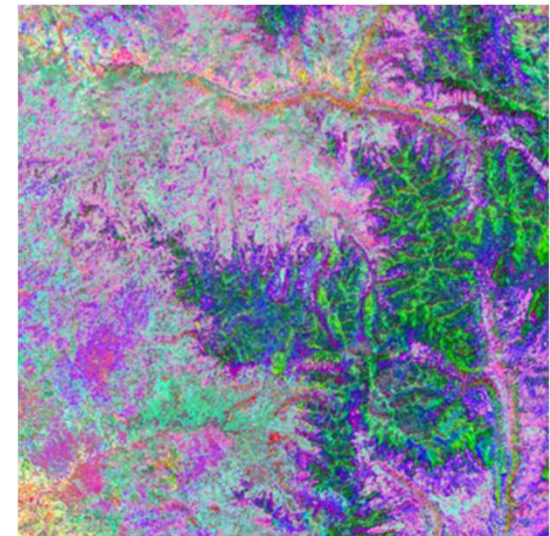
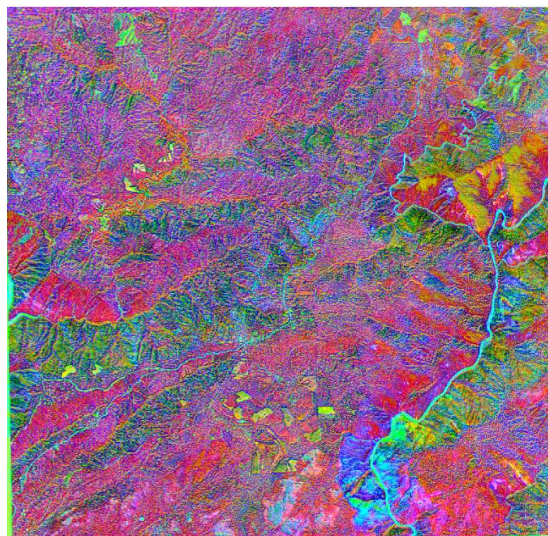
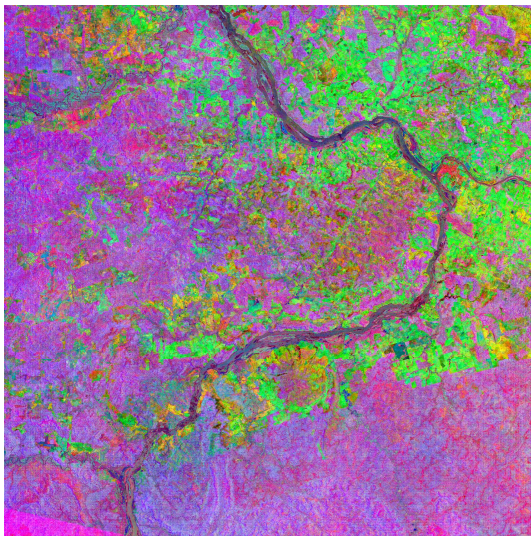
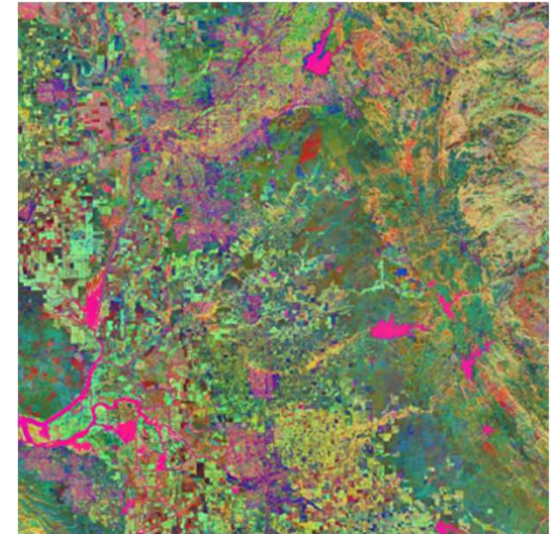
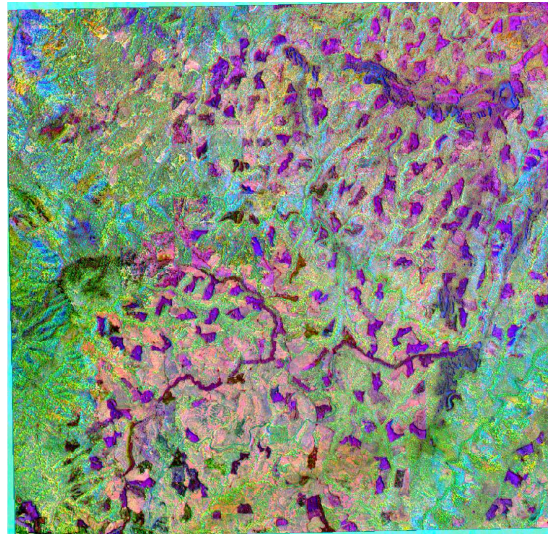
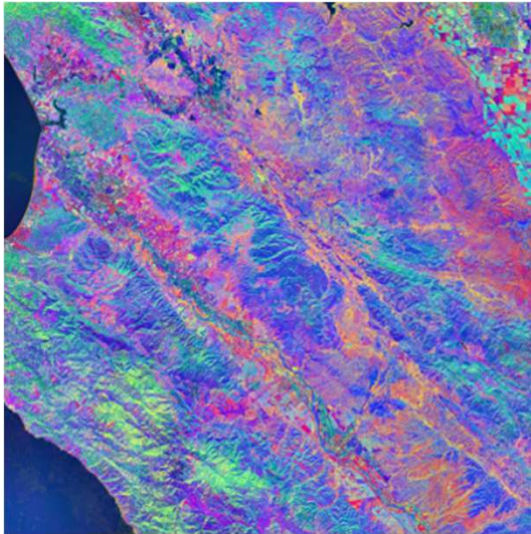


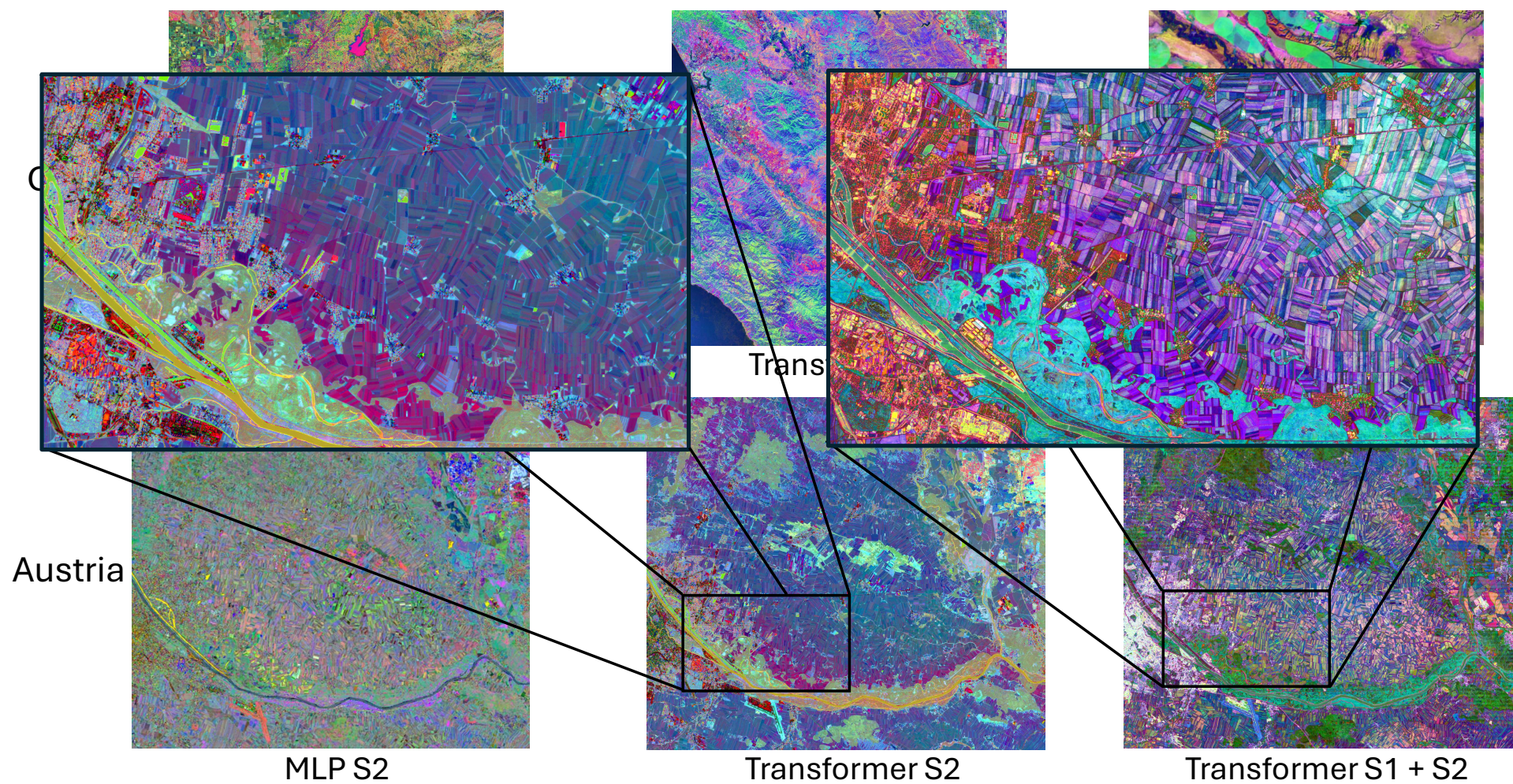
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2.Methodology

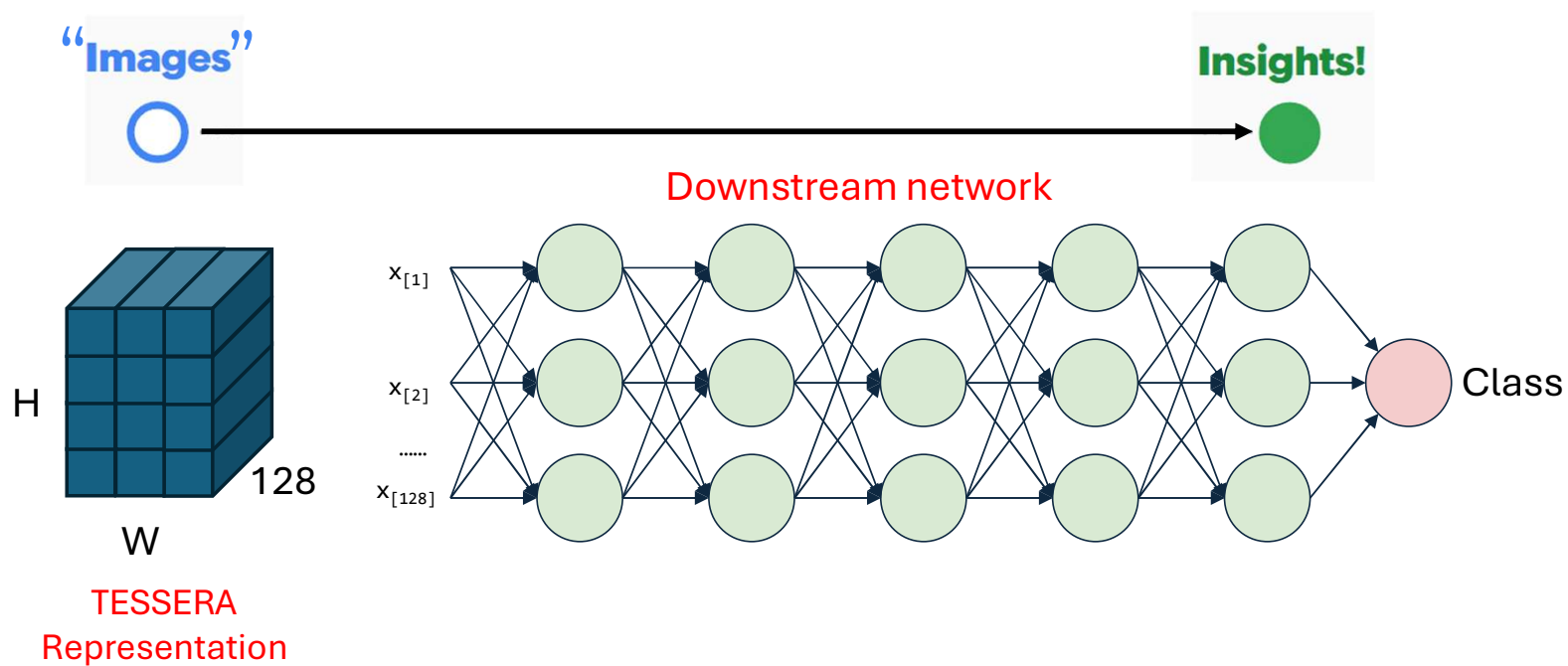
3.Result

4. Future Work

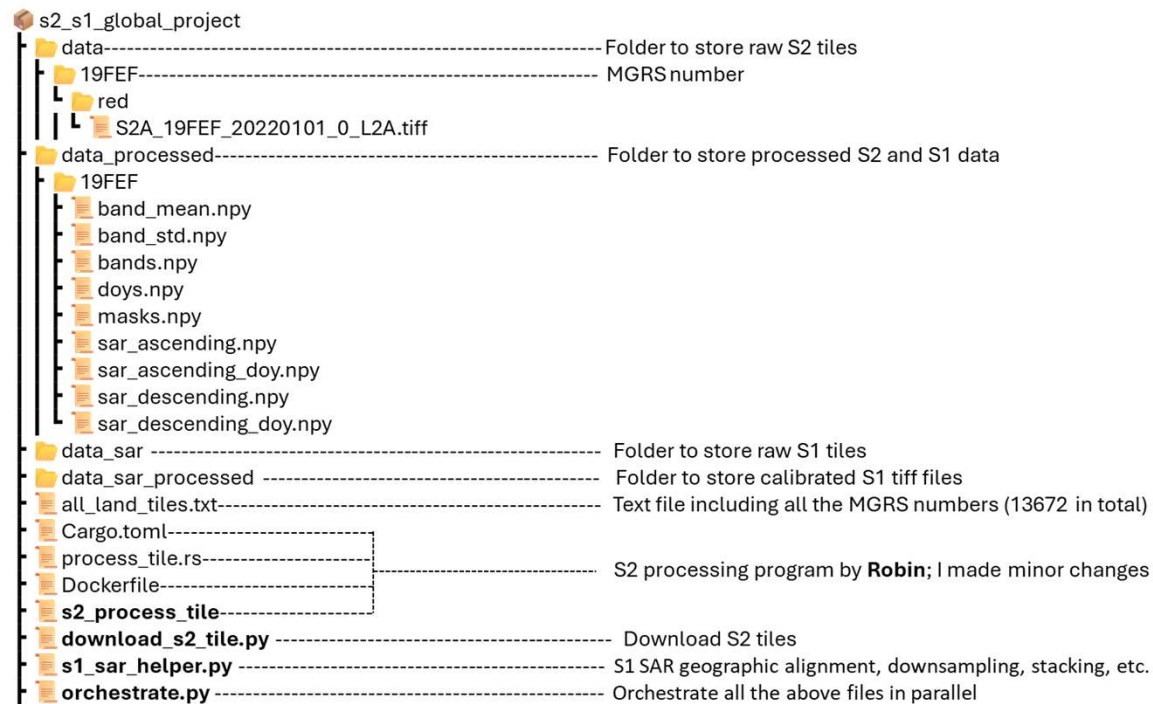




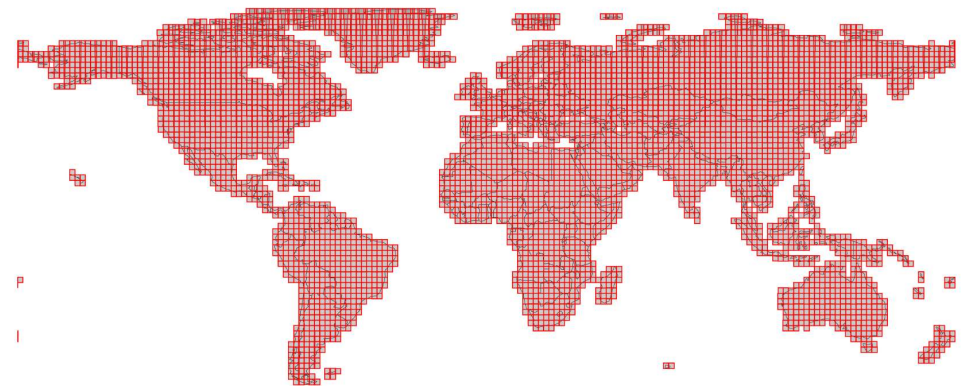
Using representations in downstream tasks



Data used to train the TESSERA Model



Trained on 3000+ MGRS tiles across the globe



Some systems aspects

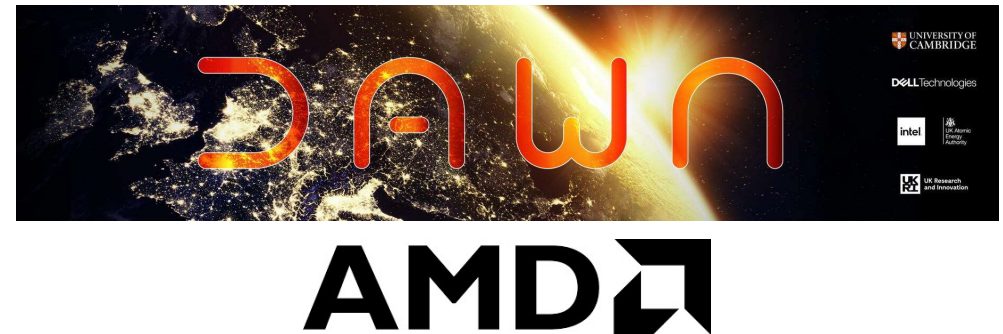
Data intensive task

- each tile is 150 GB

Need both **raw compute** and **algorithmic design**

Compute

- Dawn HPC
- AMD 8-GPU node
- 256-core 1TB RAM server
- ~300TB SSD



Algorithms

- Per-tile d-pixel creation time : **2w -> 30 min**
- Typical ROI inferencing: **10h -> 30s**

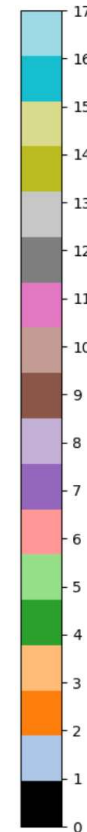
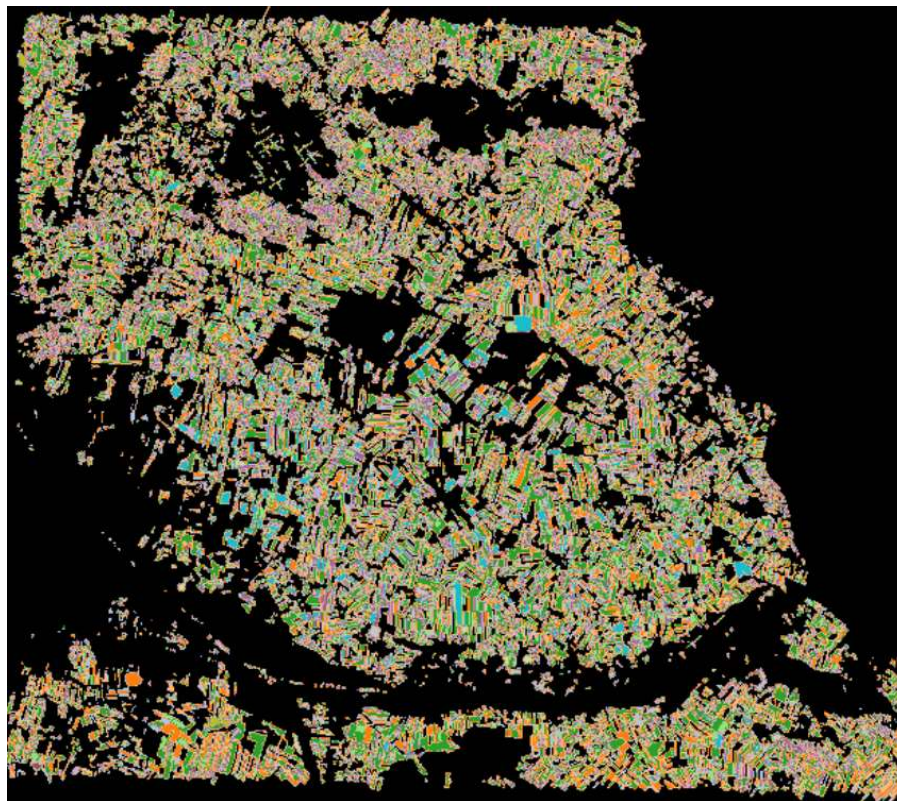


Crop Classification

Classification Task

17 classes

Austrian Crop Dataset



Name	Number of Samples	Percentage
Legume	2227	3.31%
Soy	5892	8.76%
Summer Grain	2475	3.68%
Winter Grain	24914	37.05%
Corn	6902	10.27%
Sunflower	207	0.31%
Mustard	1734	2.58%
Potato	2514	3.74%
Beet	1257	1.87%
Squash	2019	3.00%
Grapes	222	0.33%
Tree Fruit	347	0.52%
Cover Crop	1418	2.11%
Grass	2349	3.49%
Fallow	4484	6.67%
Other (Plants)	8220	12.23%
Other (Non Plants)	57	0.08%
Total	67238	100%



Crop Classification

Preliminary Results

Encoder	Region	Data	Acc	F1
RF	Austria	S2	74.39	73.10
MLP	Austria	S2	75.21	73.58
Transformer	Austria	S2	78.43	77.61
Transformer+	Austria	S2	79.48	78.77

TESSERA outperforms RF

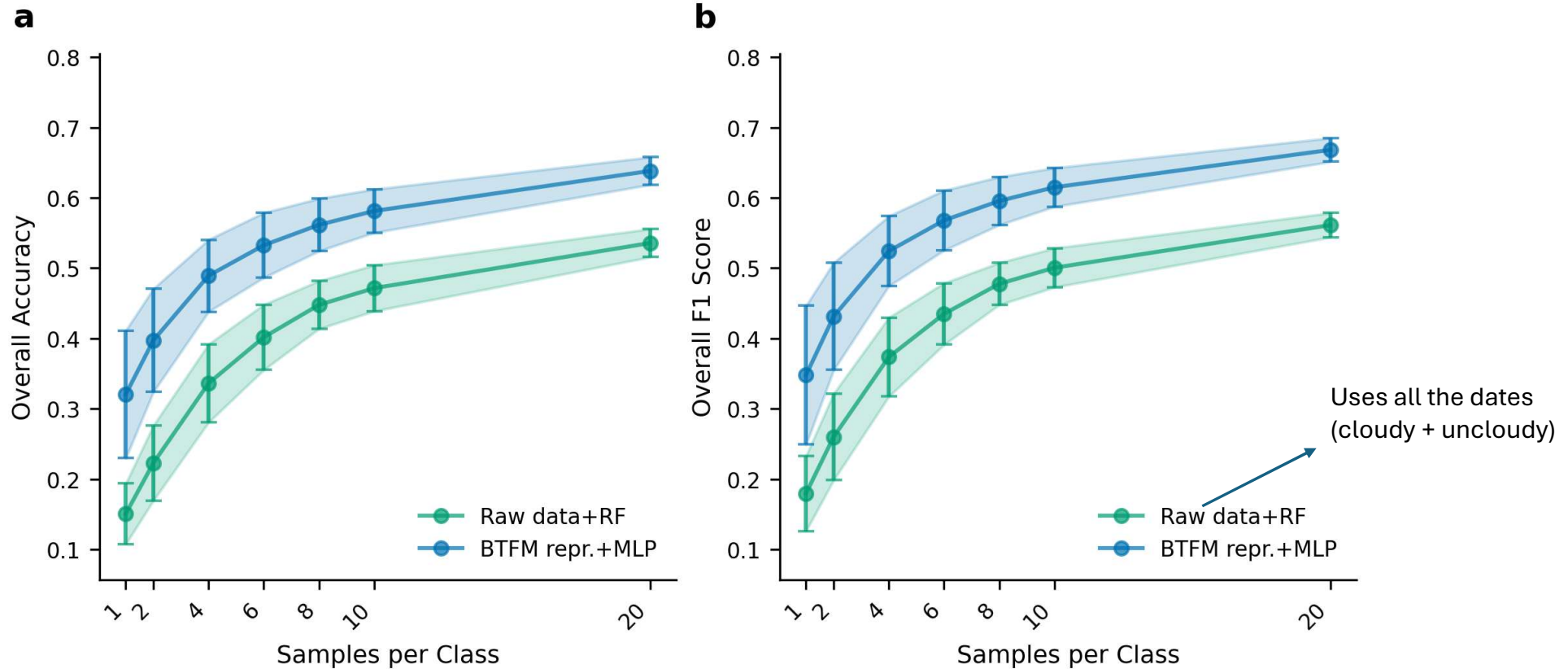
Model	Region	Data	Acc	F1
Transformer	California	S2	76.02	75.24
Transformer	Austria	S2	78.43	77.61

Representation can generalize



Crop Classification

What if we only have **very limited samples with ground truth** labels (close to real-world scenarios)



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Competition: Google Embedding Field Model

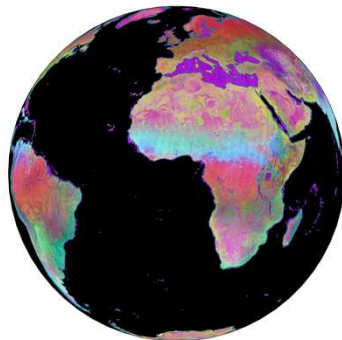
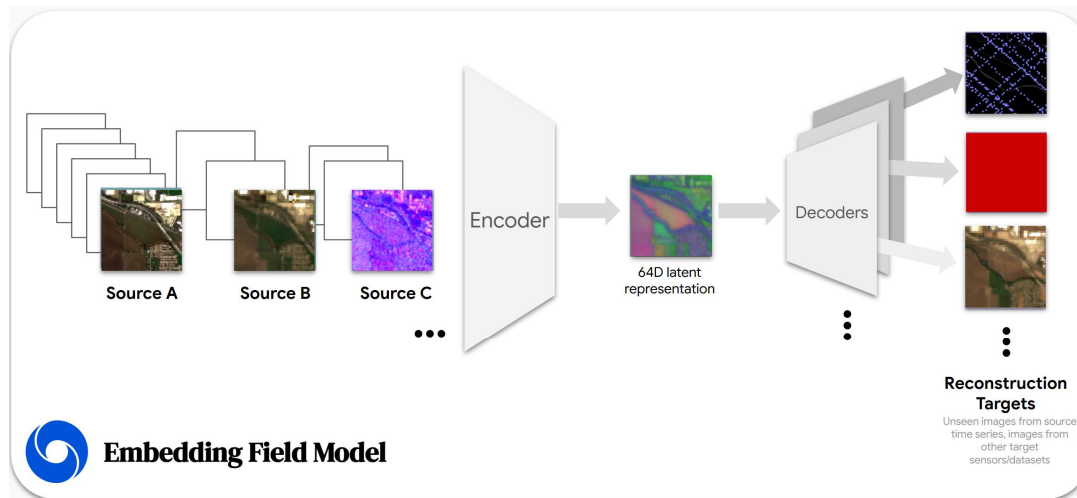


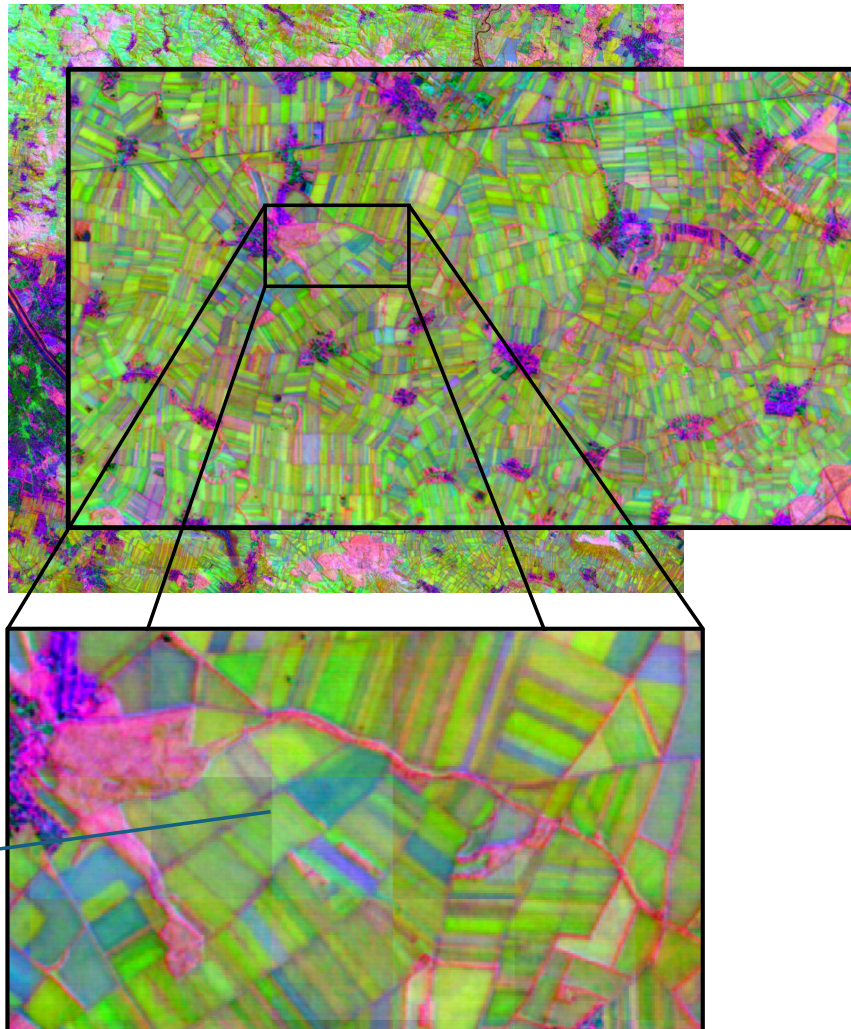
Table 1: EFM v2 data sources

Type	Dataset	Product	Bands	Resolution (m)	Usage
Optical	Sentinel-2	L1C	B2 (Blue), B3 (Green), B4 (Red), B8 (NIR), B11 (SWIR)	10, 20, 60	input, target
Optical, Thermal	Landsat-8, Landsat-9	L1C	B2 (Blue), B3 (Green), B4 (Red), B5 (NIR), B6 (SWIR), B8 (Panchromatic), B10 (Thermal)	15, 30, 100	input, target
C-band SAR	Sentinel-1A, Sentinel-1B	GRD	VV, VH, HH, HV, angle	10	input, target
L-band SAR	ALOS PALSAR ScanSAR	Level 2.2	HH, HV, lin	25	target
Elevation	Copernicus DEM	GLO-30	DEM (elevation)	30	target
LiDAR	GEDI	L2A	Relative height metrics (rh*)	25	target
Climate	ERA5-Land	Monthly aggregates	total precipitation (sum, min, max), air temperature 2m (and min, max), dew-point temperature 2m (and min, max), surface pressure (and min, max)	11132	target
Gravity fields	GRACE	Monthly mass grids	equivalent liquid water thickness	11132	target (@50%)
Land cover	National Land Cover Database	NLCD 2019, 2021	landcover	N/A	target (@50%)
Text	Wikipedia	geocoded articles	text embeddings	N/A	target
Text	GBIF	Research-grade obs	text embeddings (class, genus, and species)	N/A	target



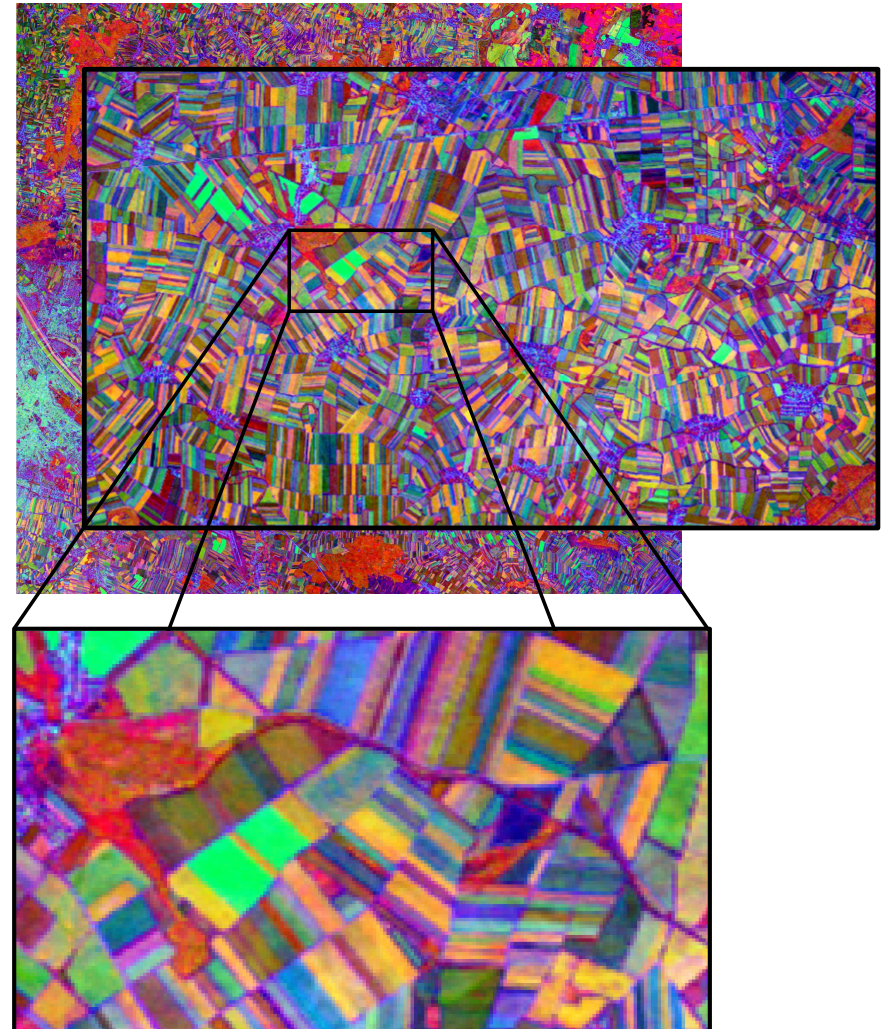
a)

EFM



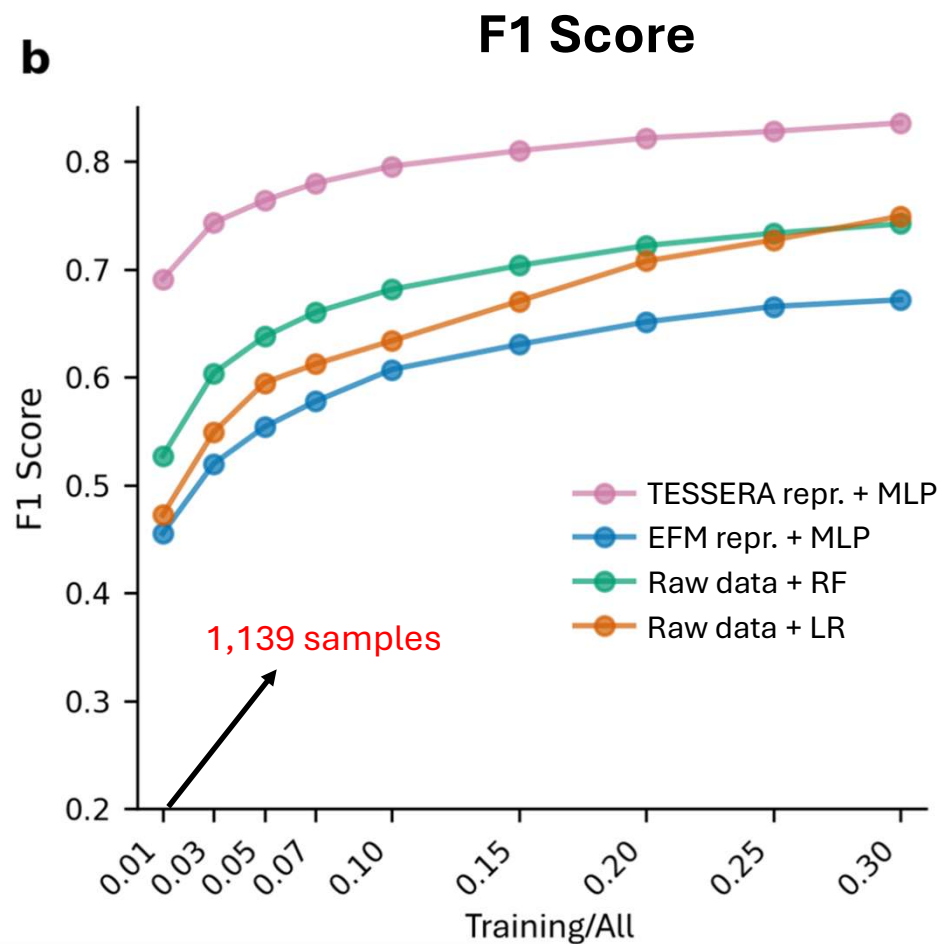
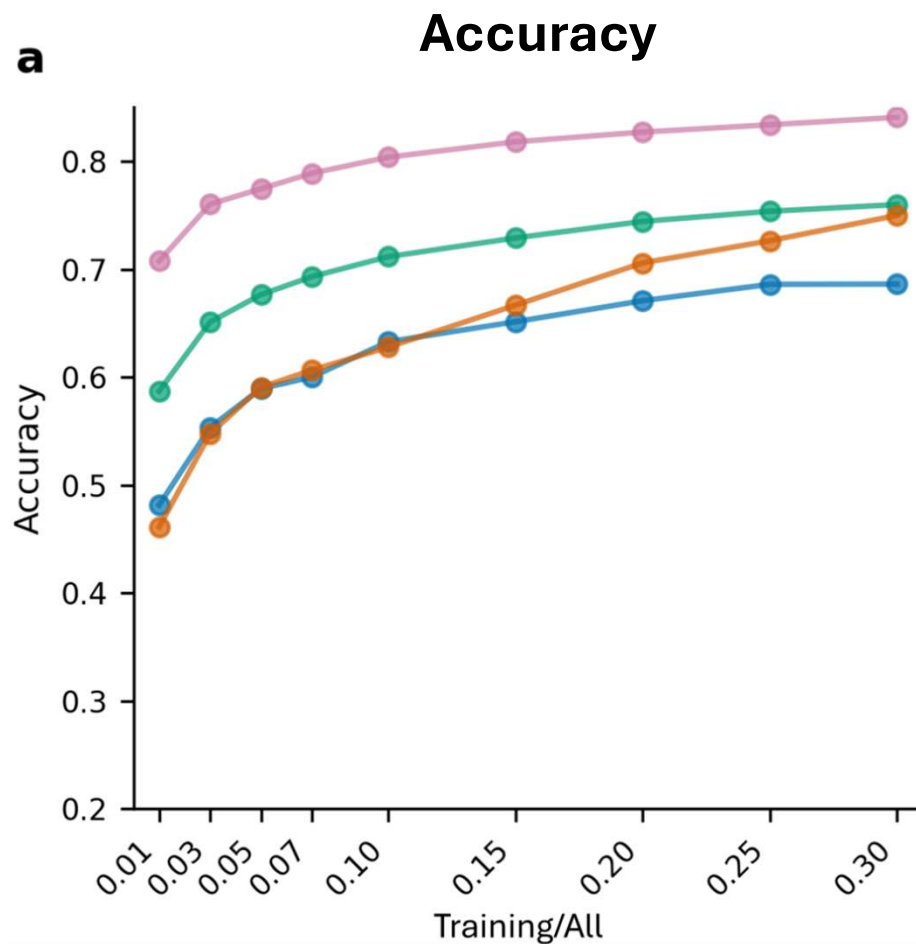
b)

TESSERA

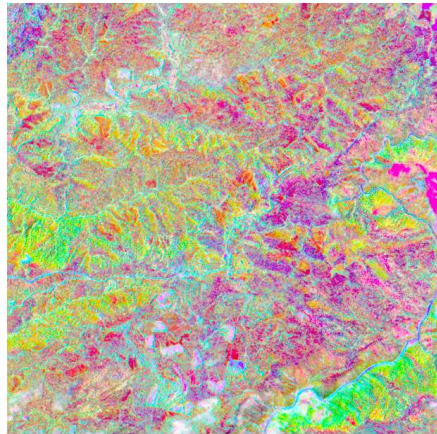


Tiling artifact

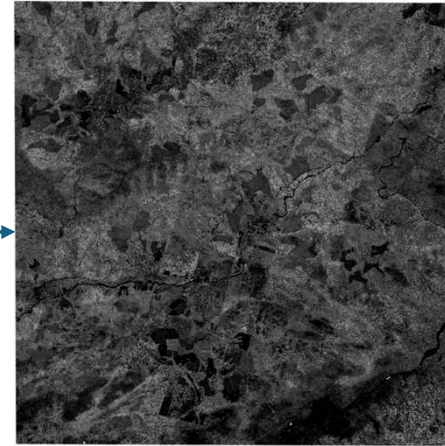
Crop Classification



Predicting Canopy Height

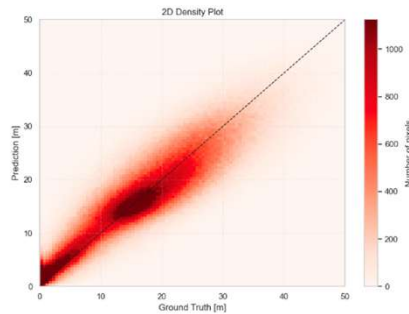


Downstream
Model
(UNet)

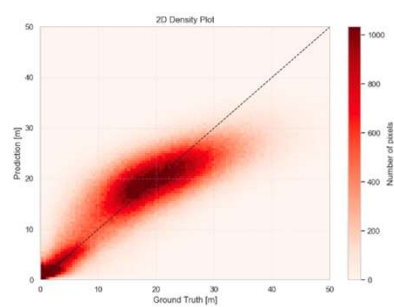


Representations

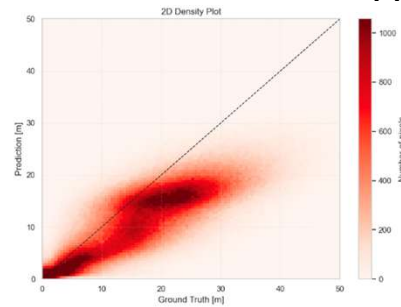
Canopy Height from
Airborne LiDAR



TESSERA
 $R^2 = 0.78$
MAE = 3.7m



EFM
 $R^2 = 0.68$
MAE = 4.4m



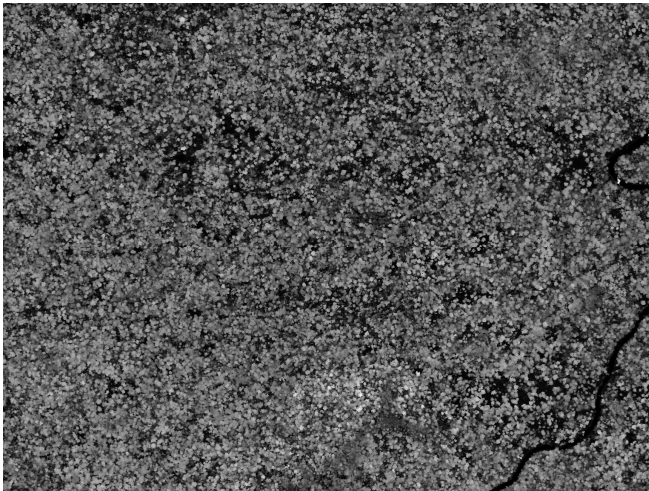
Sentinel-2 Medians
 $R^2 = 0.34$
MAE = 6.5m

Forest in Northern California

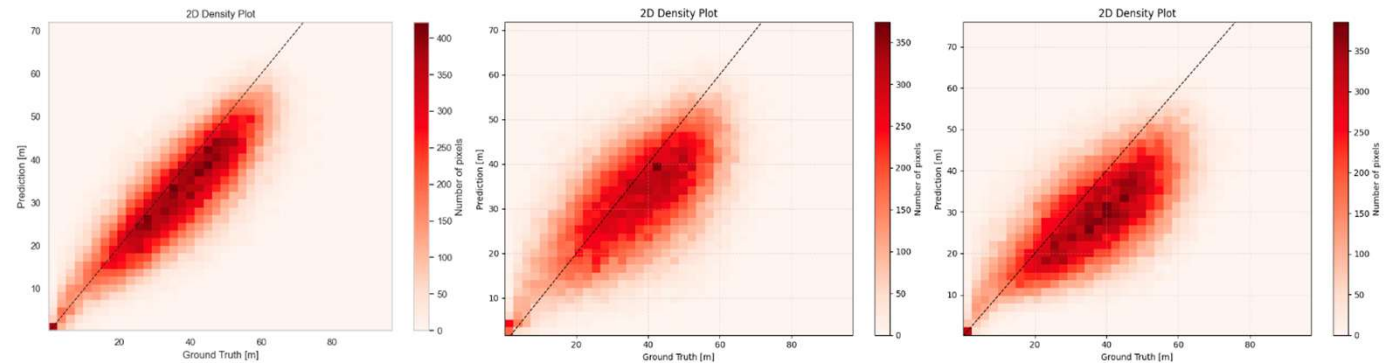


Predicting Canopy Height

Borneo Forest



Different inputs, different predictive power

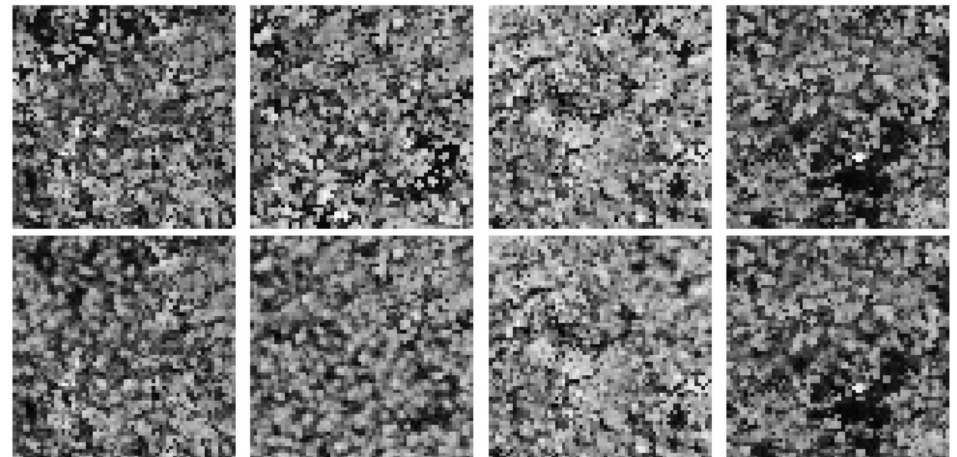


TESSERA
 $R^2 = 0.55$

EFM
 $R^2 = 0.25$

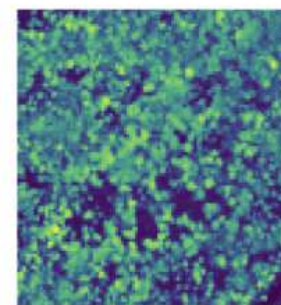
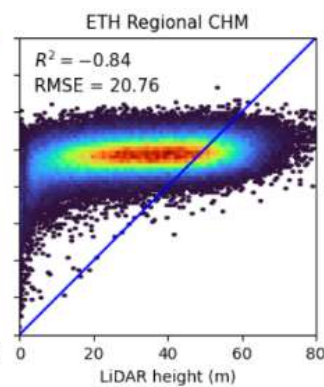
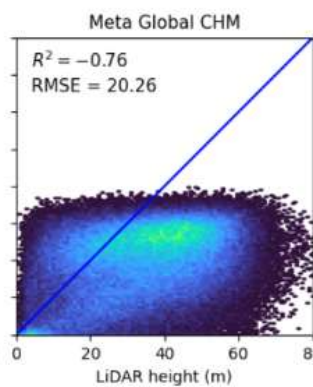
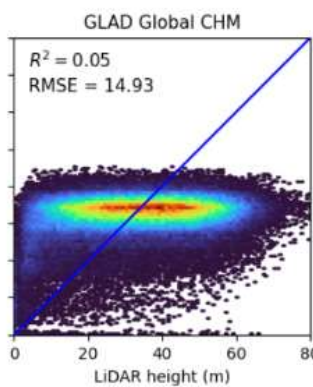
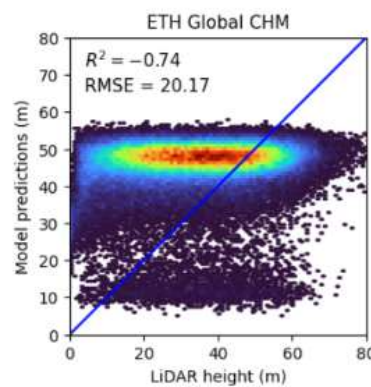
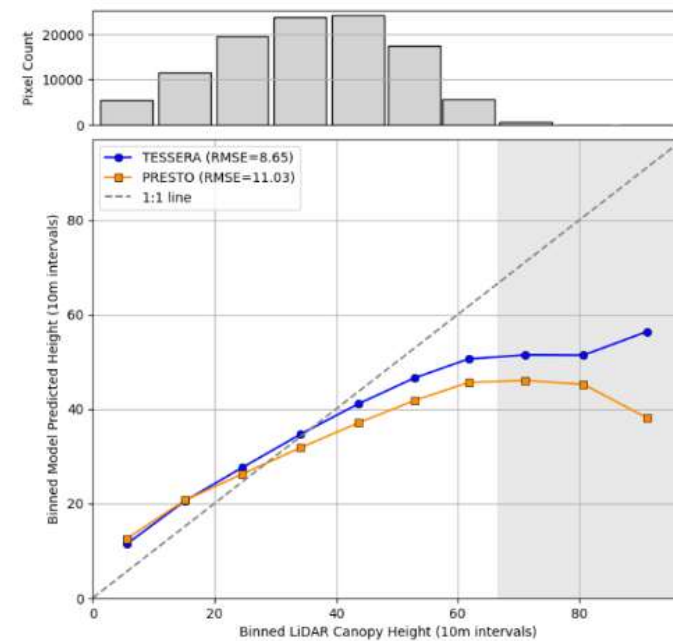
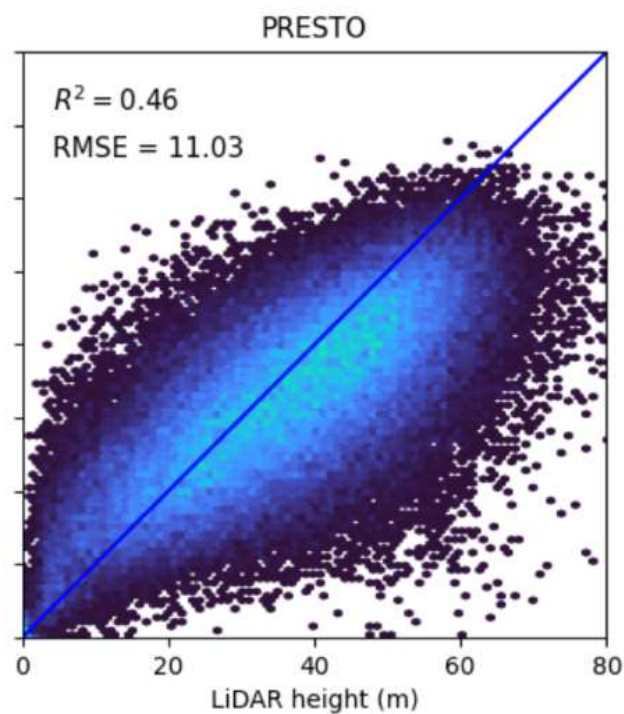
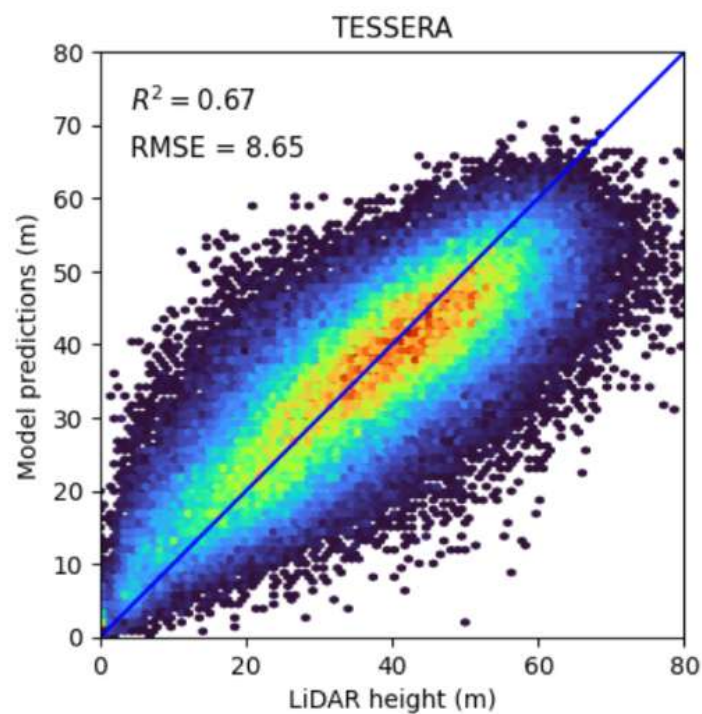
Sentinel-2 Medians
 $R^2 = 0.17$

Ground Truth

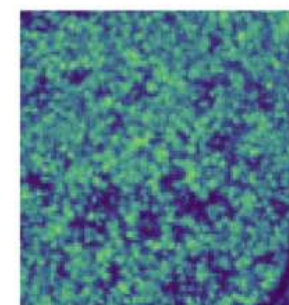


Predicted





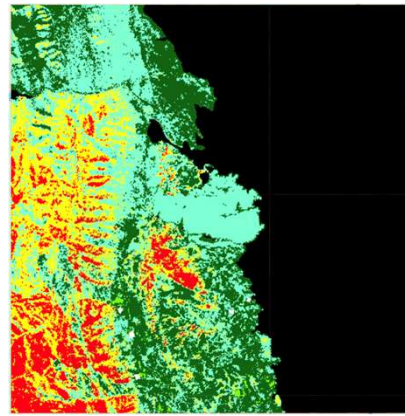
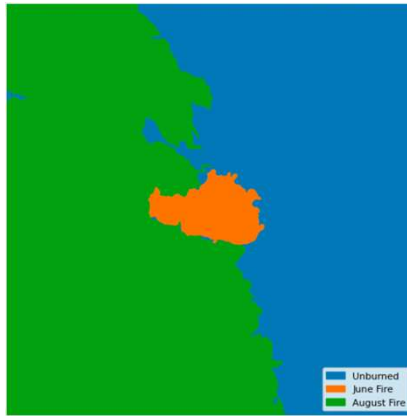
True Canopy Height



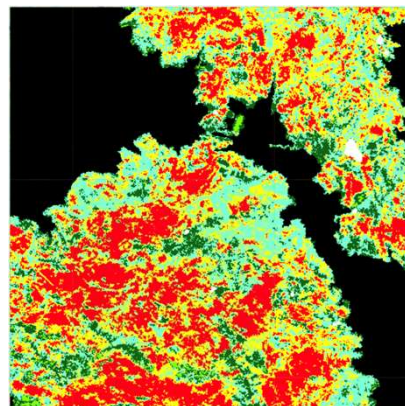
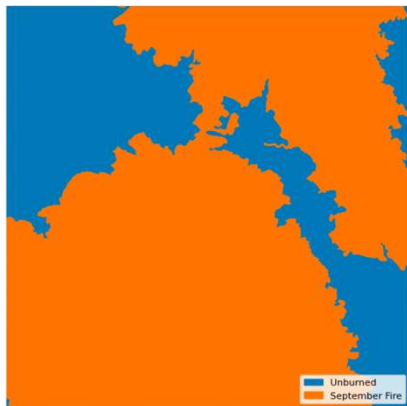
TESSERA Predicted Canopy Height

Detecting Disturbances: *Fires in California*

Burned Area 1

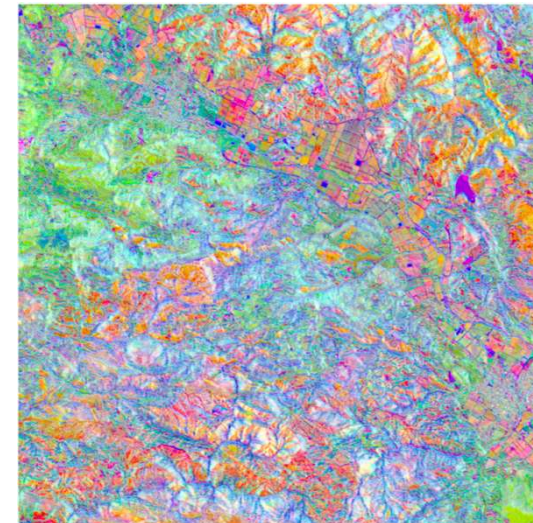
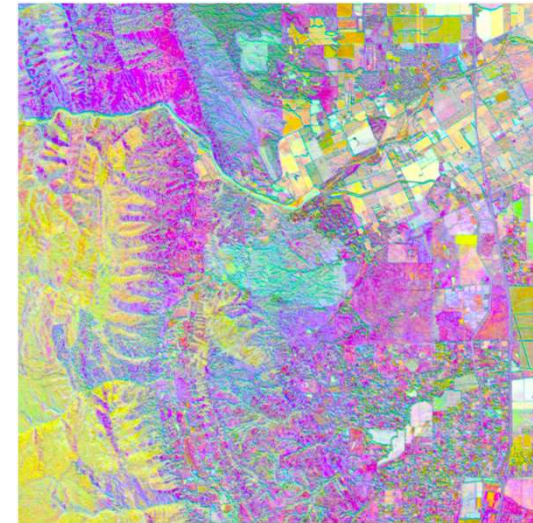


Burned Area 2



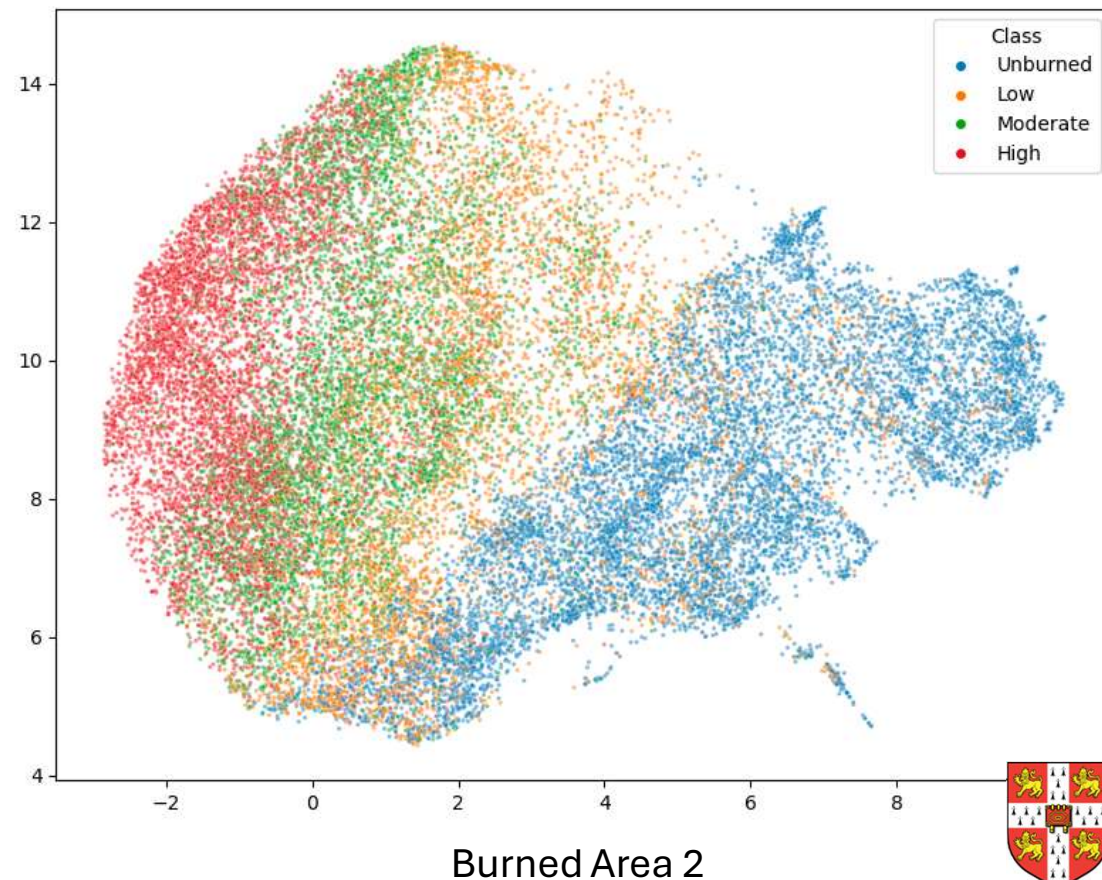
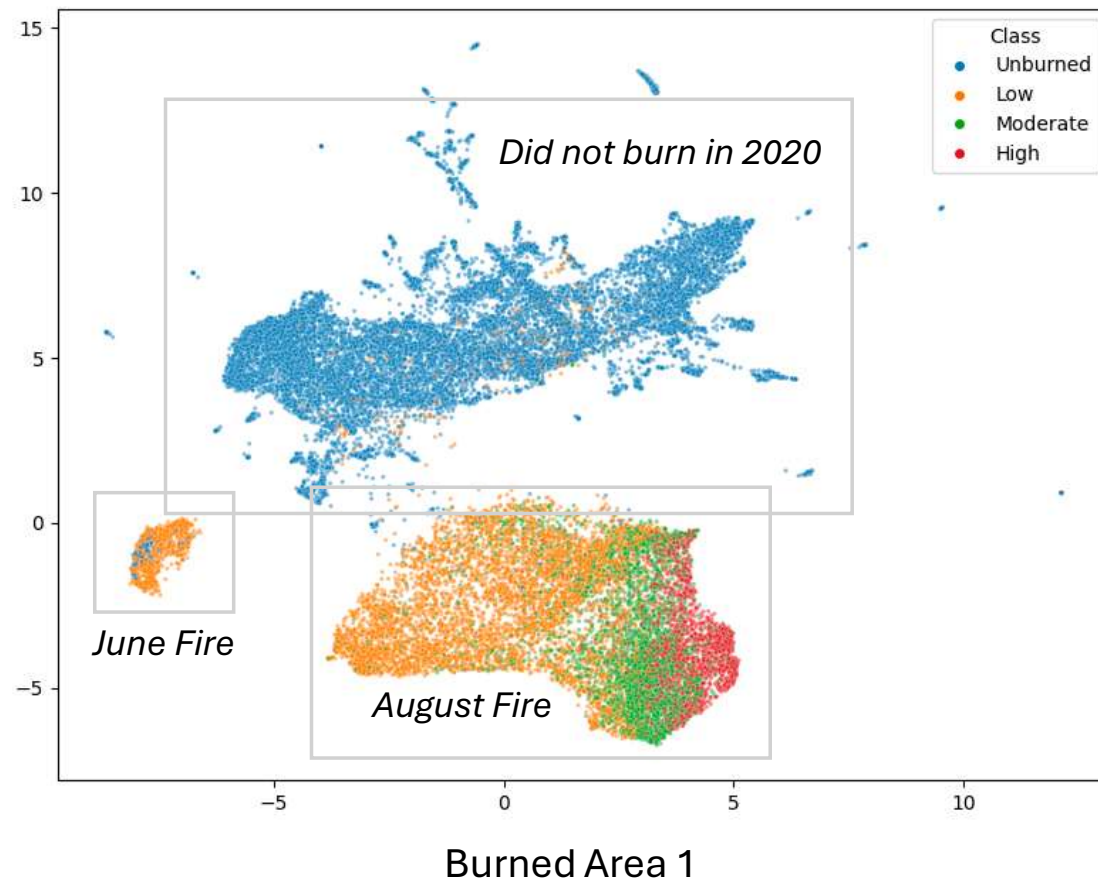
Fires in 2020

Fire Severity

TESSERA
Representations
For 2020

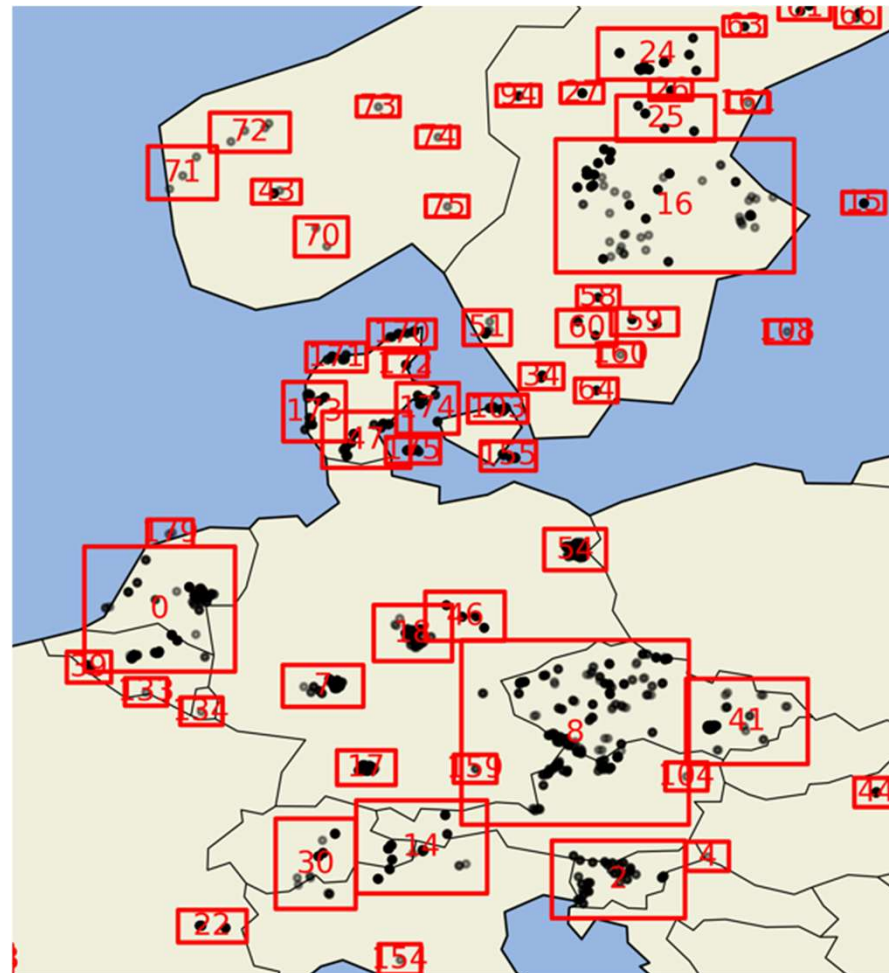
Detecting Disturbances: *Fires in California*

UMAP (Dimensionality Reduction 128 -> 2)



Fungal Biodiversity

Ground truth (Mycorrhizal biodiversity from SPUN)

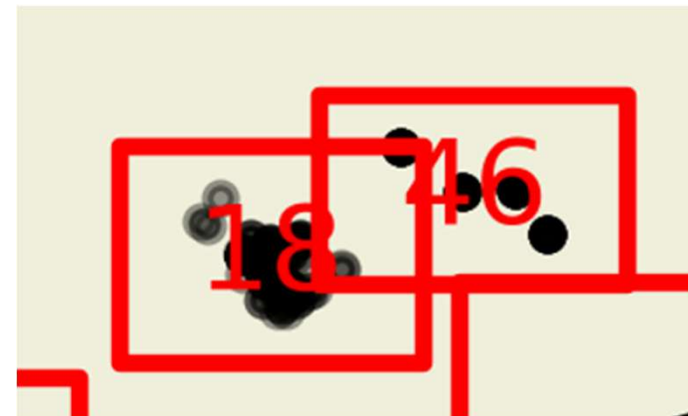


Fungal Biodiversity

Steps:

- From lat-lon data, get max,min lat-lon, create bounding box (**this is the ROI**)
- Download S1/S2 data, **create d-pixels**
- **Inference** with checkpoint, get **representation**
- Match representations to ground truth sample (x-y pair)

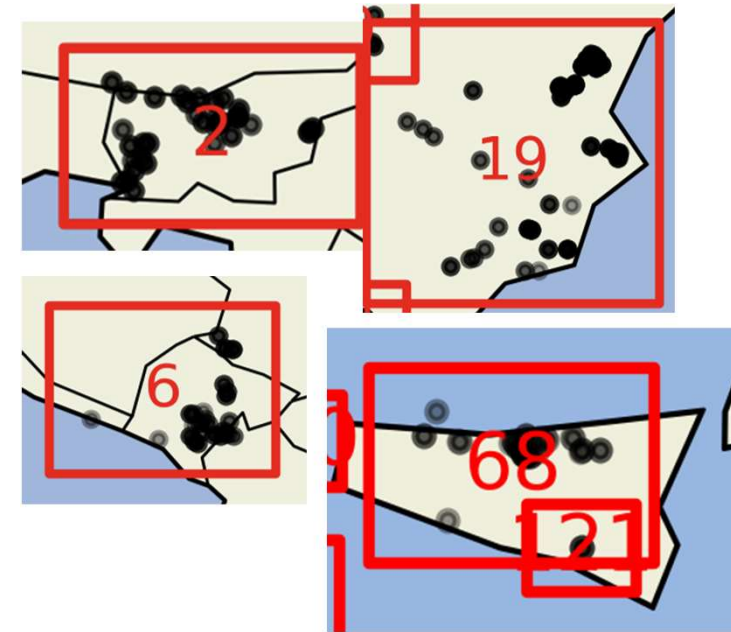
Apply random forest/xgboost, **train/test**



Fungal Biodiversity

Early Results

	Germany	Slovenia + Kosovo + Sicily	Estonia + Latvia
R^2	~0.65	~0.5	~0.2

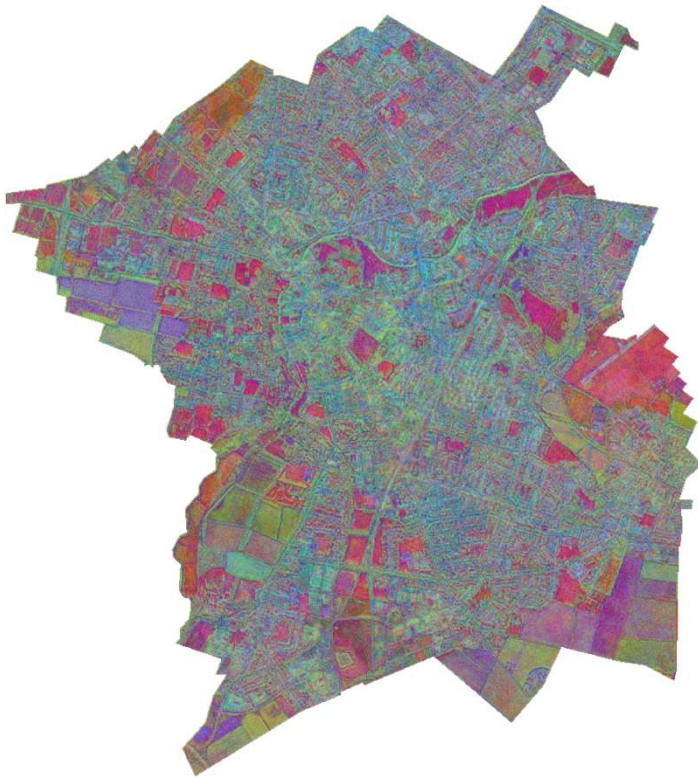


For Estonia, possibly some underlying ecological reasons
(Seasonal waterlogging messing with satellite reflectances)

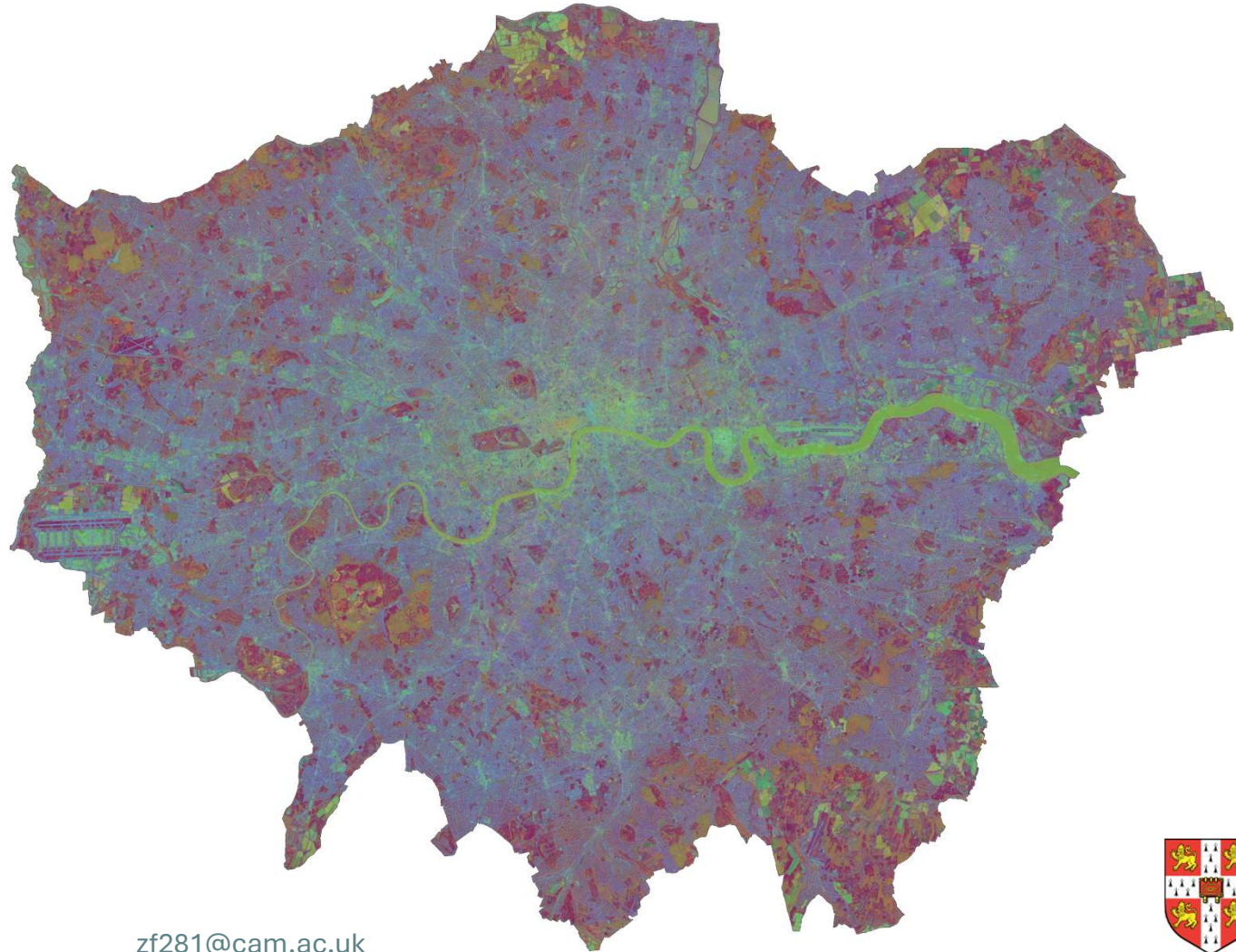


Land Change Detection

Year: 2017

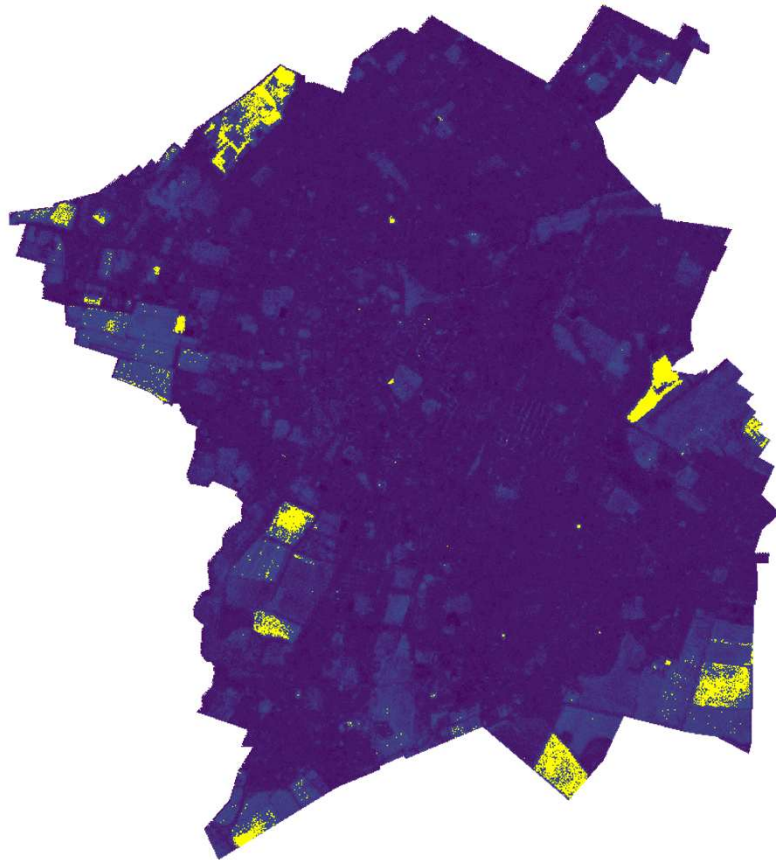


Year: 2017



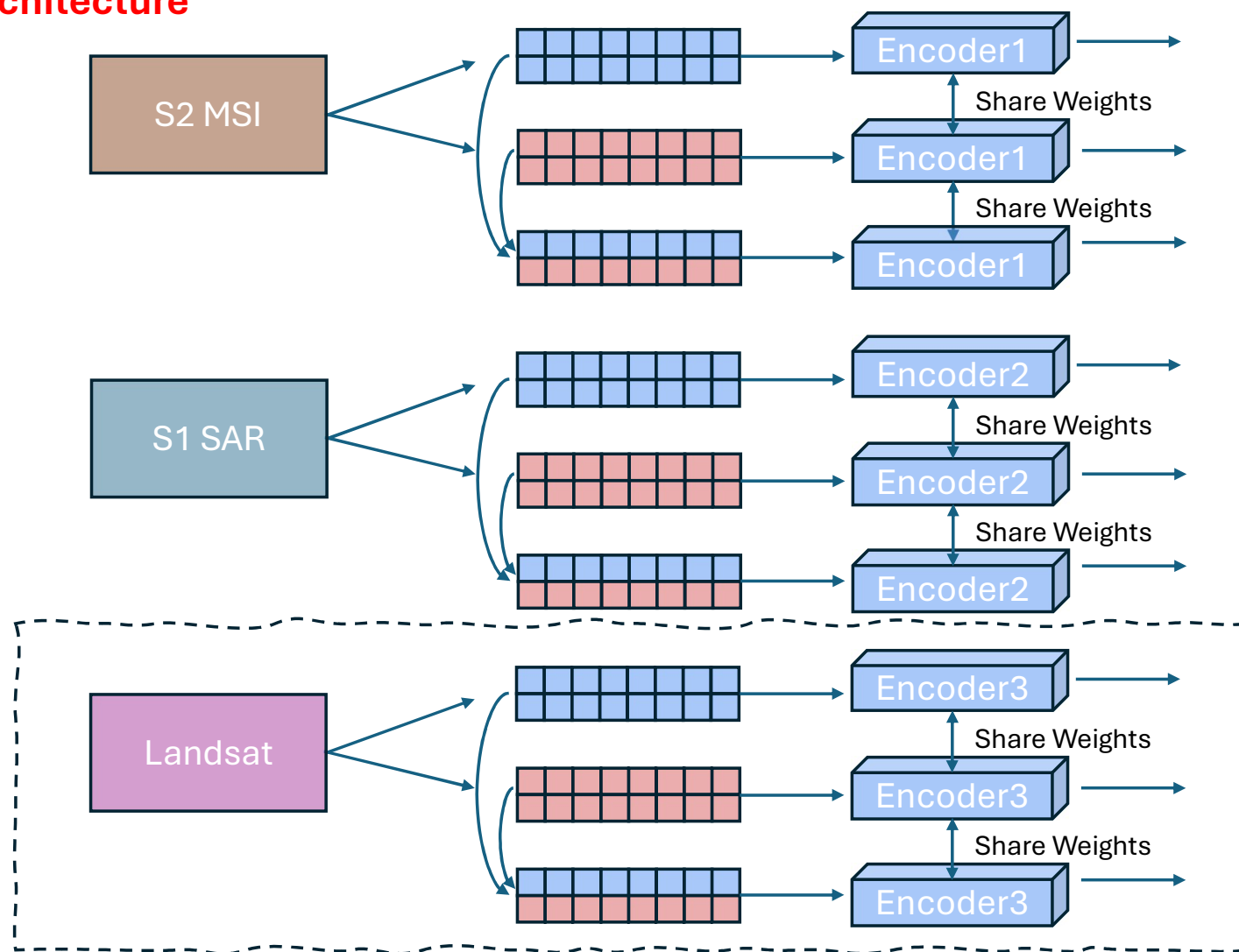
Land Change Detection

2017-2018



2017-2018



Model Architecture

Barlow Twins

Barlow Siblings

2016-2024

1973-2024

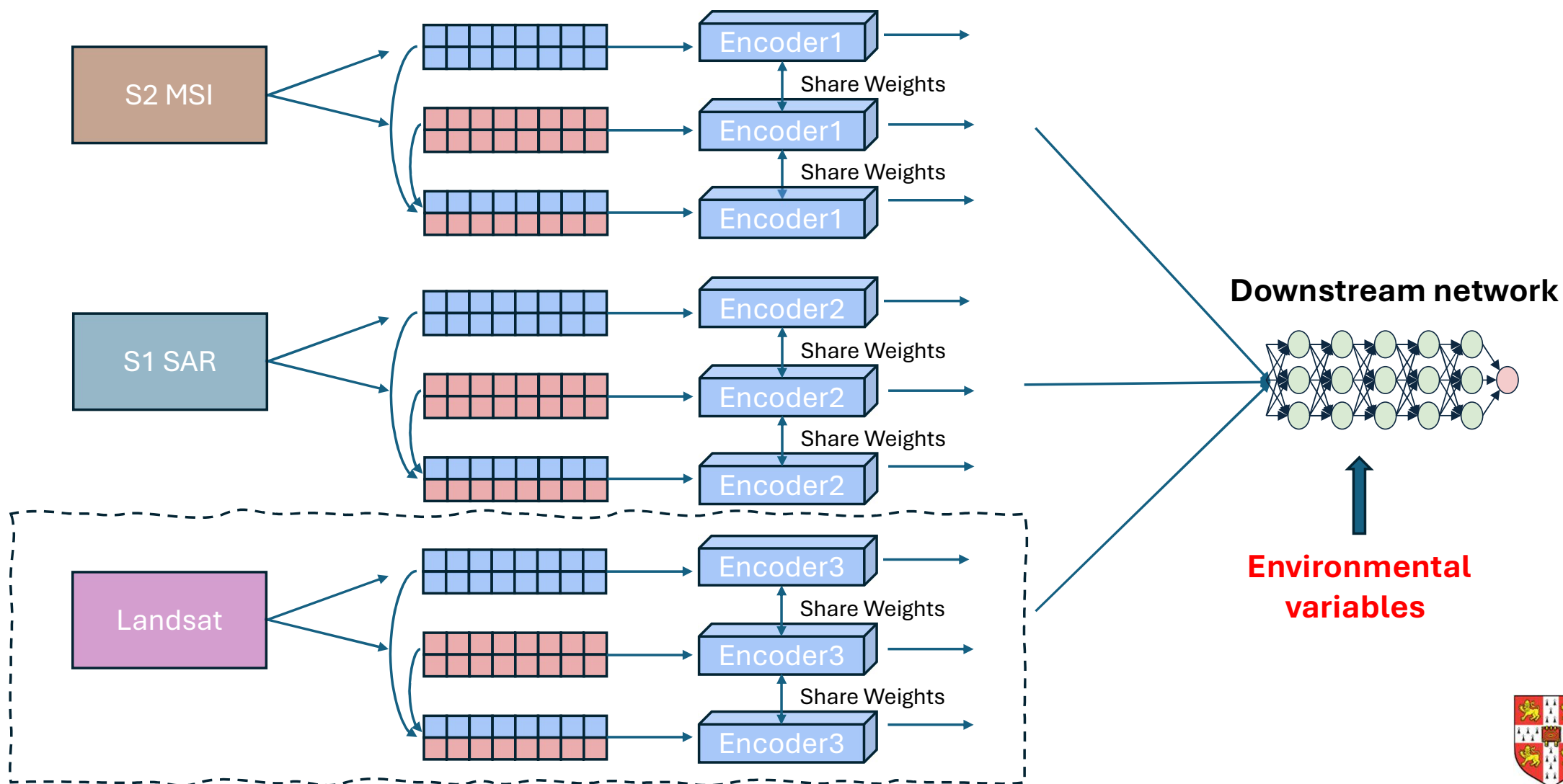


1.Introduction

2.Methodology

3.Result

4. Future Work



More downstream tasks

Land segmentation

Crop yield analysis

Counterfactual pixel matching

Using representations to understand land use change

Species distribution modelling/Robust Area of Habitat (AoH) classifiers for plant biodiversity

Comparing JRC degradation maps with the representations estimate of degradation

Global dynamic forest carbon maps

Local carbon and disturbance mapping

Global dynamic habitat maps

...



Inter-disciplinary team

Computer Science

- AI
- Systems
- Computer vision

Frank Feng
Jovana Knezevic
Robin Young
Sadiq Jaffer
Andrew Blake
S. Keshav
Anil Madhavapeddy

Remote sensing

- Satellite data
- Radiative transfer models

Maddy Lisaius
Frank Feng
Clement Atzberger

Agriculture/Ecology

- Forest processes
- Crop behaviour

Maddy Lisaius
Clement Atberger
David Coomes



Conclusions

- SSL extracts useful insights from unlabeled EO data
- Barlow siblings approach helps improve diverse downstream tasks over SOTA
- We are seeking collaborators



TESSERA

Version:0.9

Github
Homepage



2017-2024 Global Representation Map

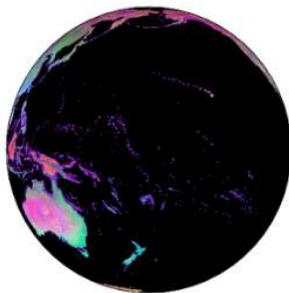
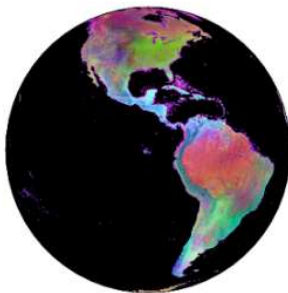
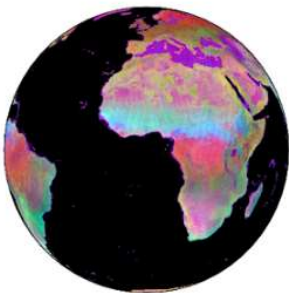
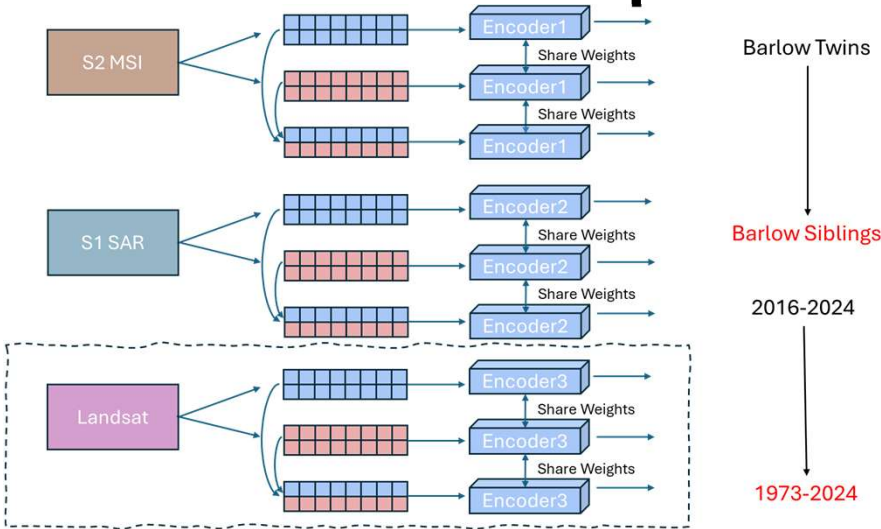


Image from Google



- s2_s1_global_project
 - data - Folder to store raw S2 tiles
 - 19FEF - MGRS number
 - red
 - S2A_19FEF_20220101_0_L2A.tiff
- data_processed - Folder to store processed S2 and S1 data
 - 19FEF
 - band_mean.npy
 - band_std.npy
 - bands.npy
 - days.npy
 - masks.npy
 - sar_descending.npy
 - sar_descending_doy.npy
 - sar_descending_doy.npy
- data_sar - Folder to store raw S1 tiles
 - data_sar_processed - Folder to store calibrated S1 tiff files
- all_tand_tiles.txt - Text file including all the MGRS numbers (13672 in total)
- Cargo.toml
- process_tile.rs - S2 processing program by Robin; I made minor changes
- Dockerfile
- s2_process_tile - S2 processing program by Robin; I made minor changes
- download_s2_tile.py - Download S2 tiles
- s1_sar_helper.py - S1 SAR geographic alignment, downsampling, stacking, etc.
- orchestrate.py - Orchestrate all the above files in parallel

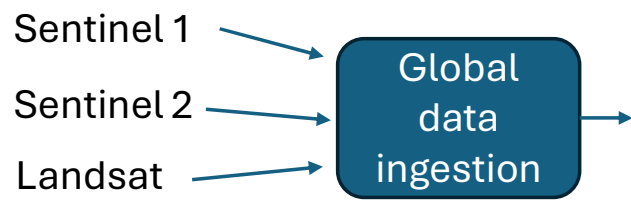


zf281@cam.ac.uk

TESSERA process

Sentinel 1
Sentinel 2
Landsat

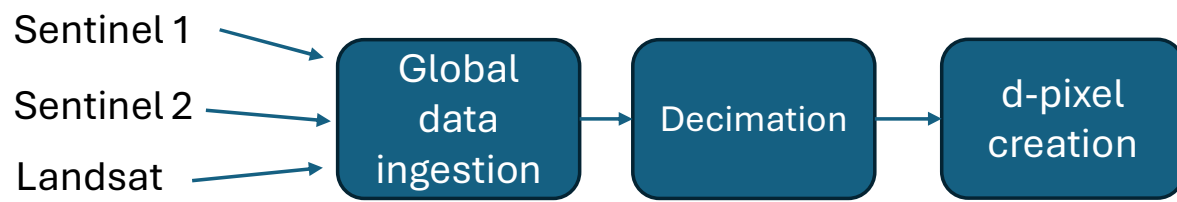
Pre-training



Pre-training

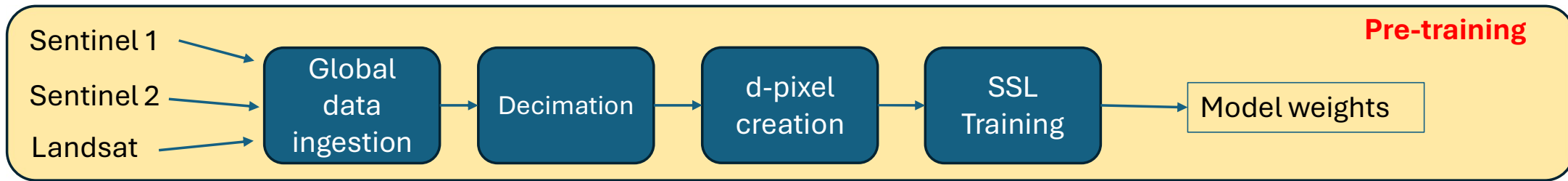


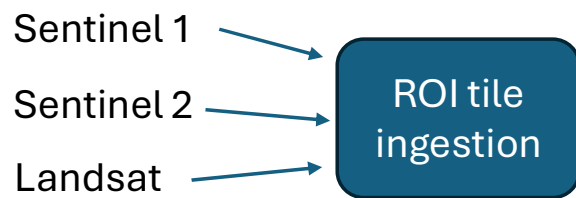
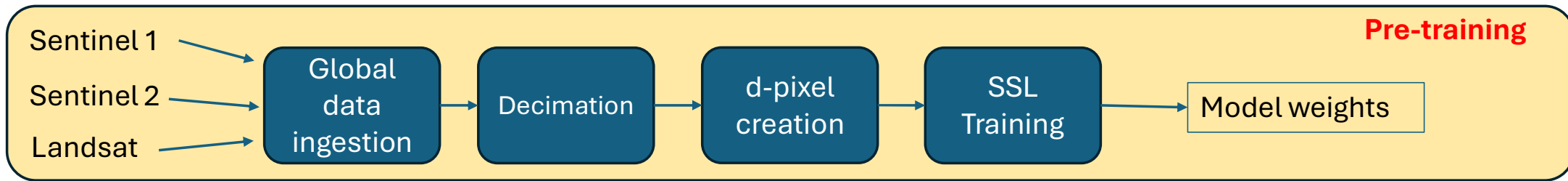
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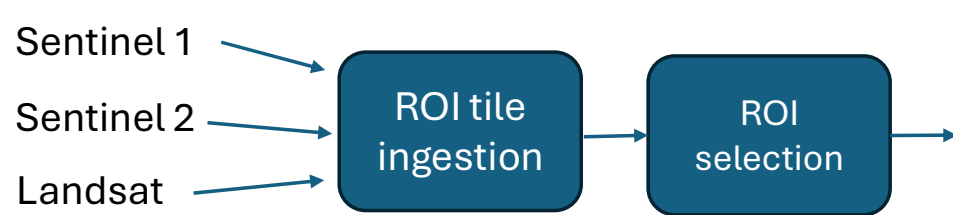
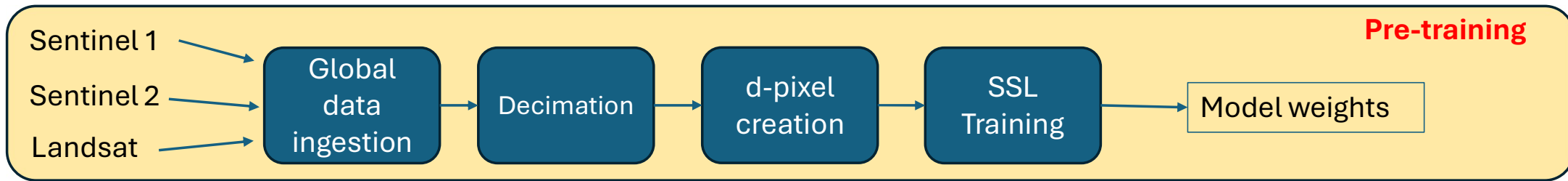


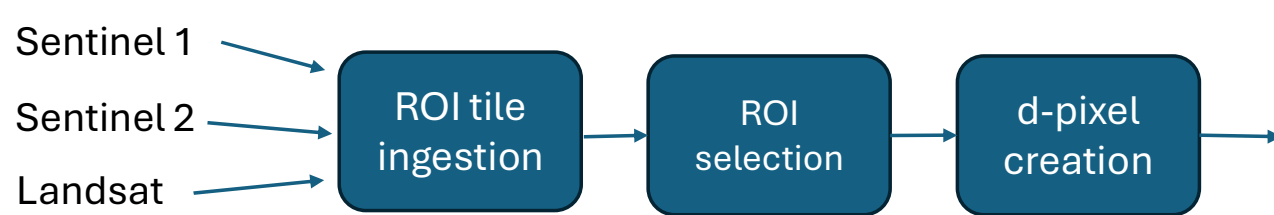
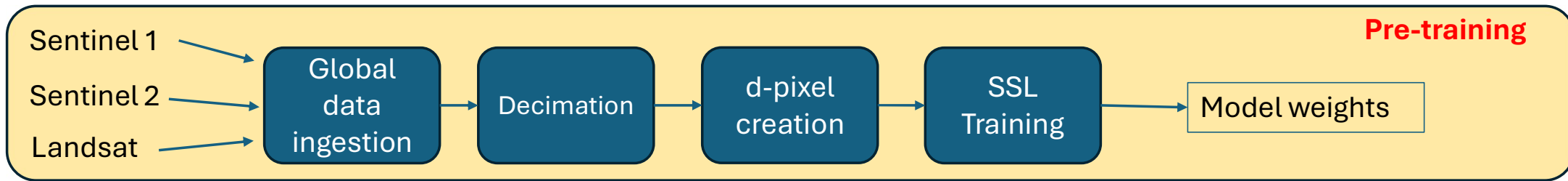
Pre-training

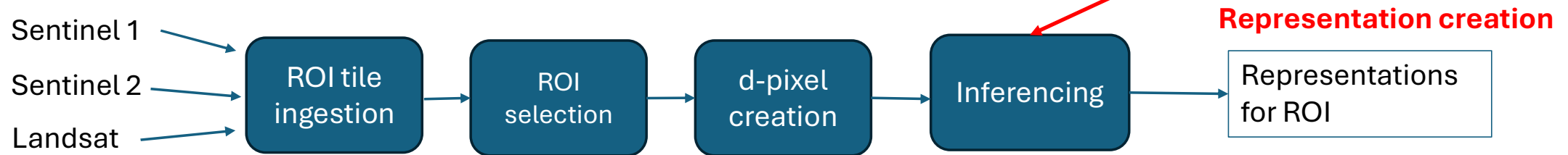
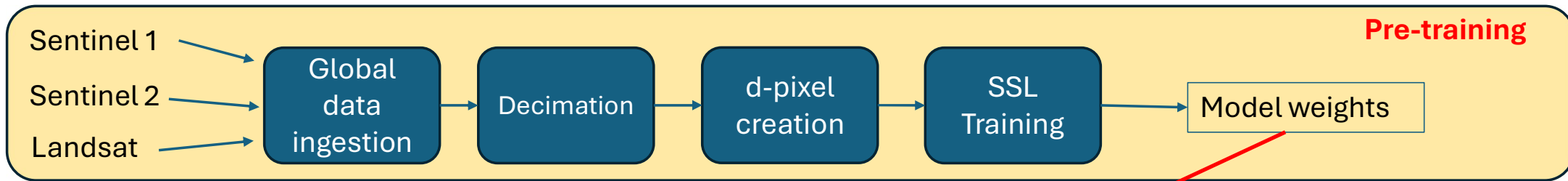


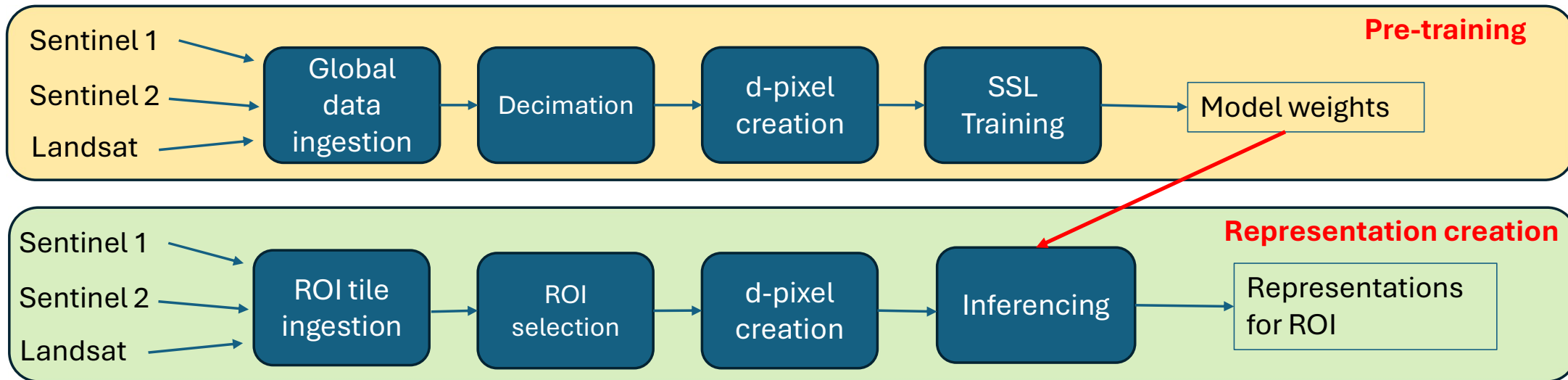


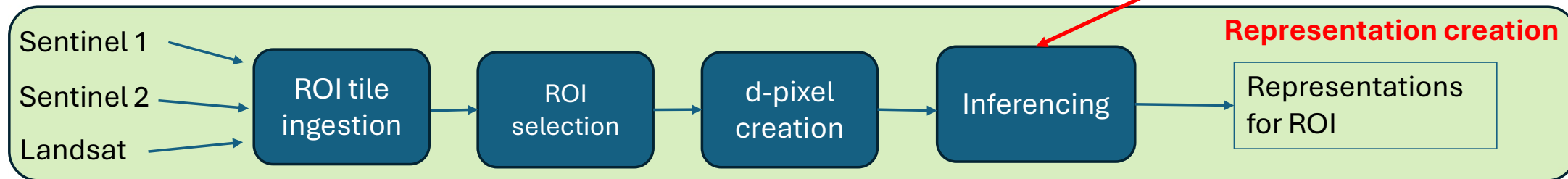
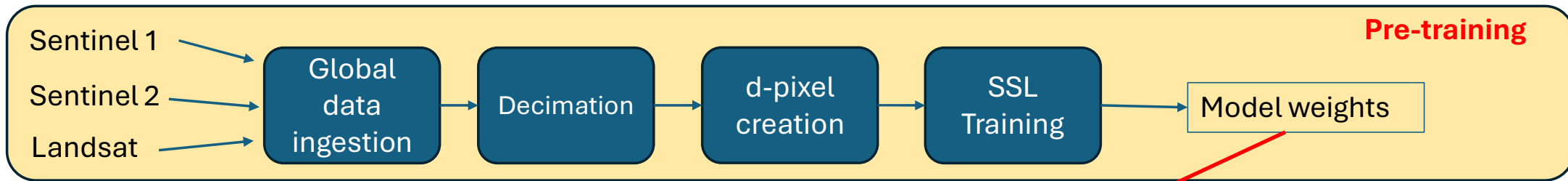






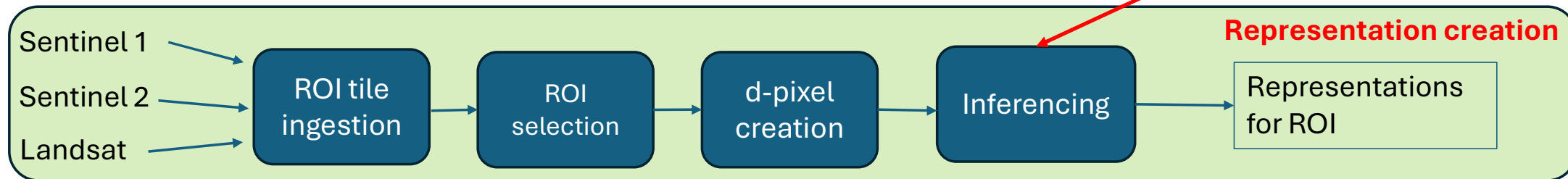
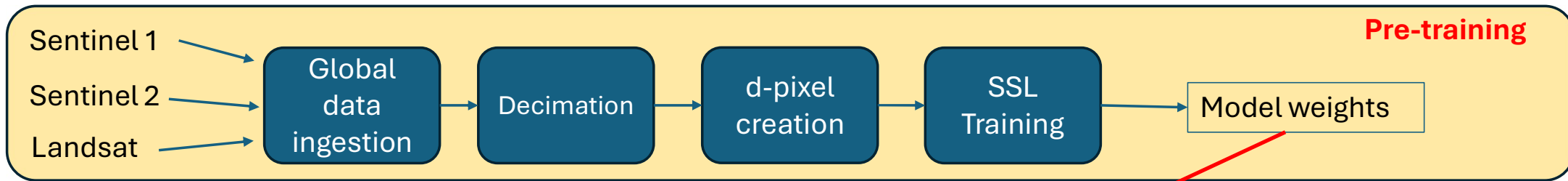






Fine-tuning

Downstream task

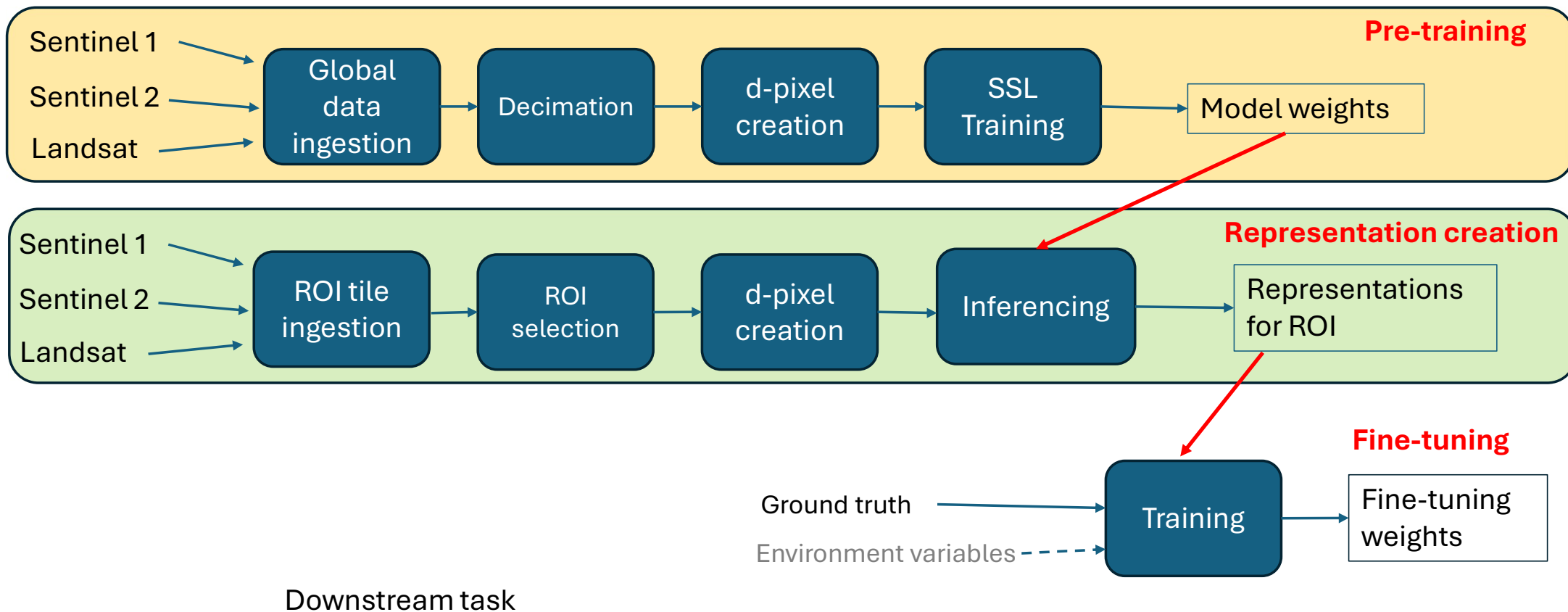


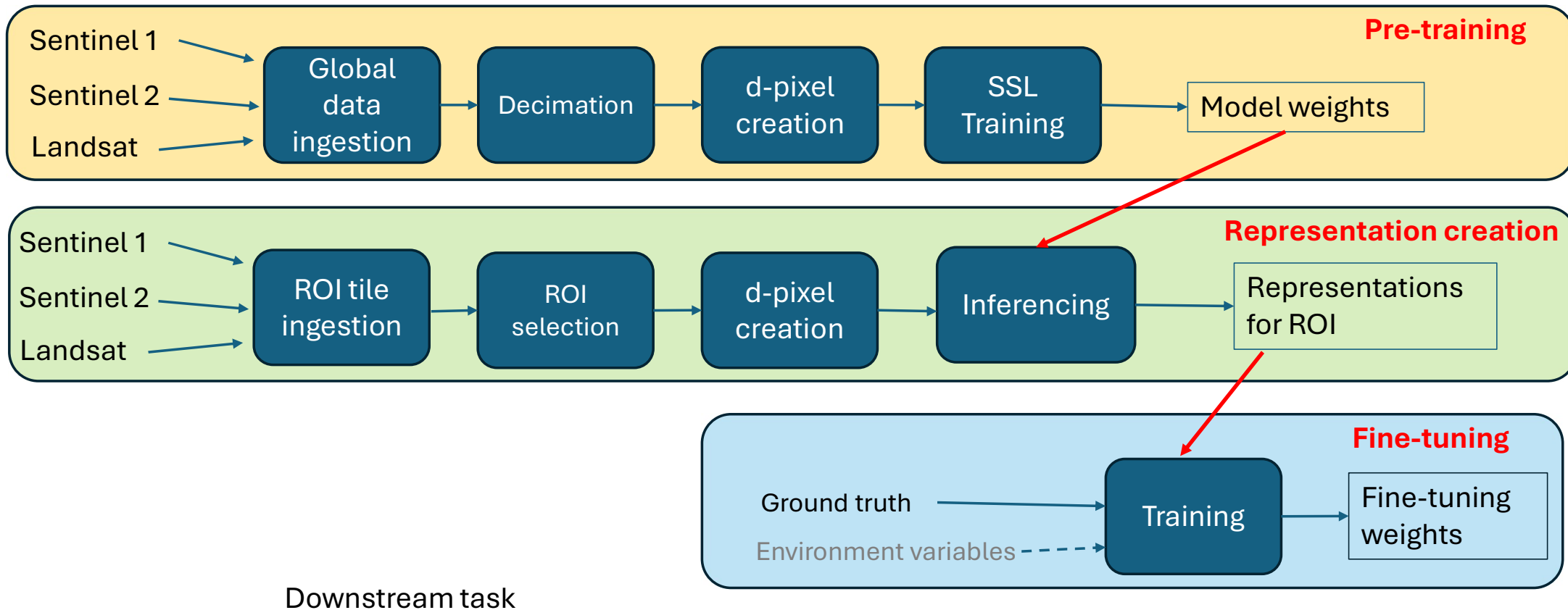
Fine-tuning

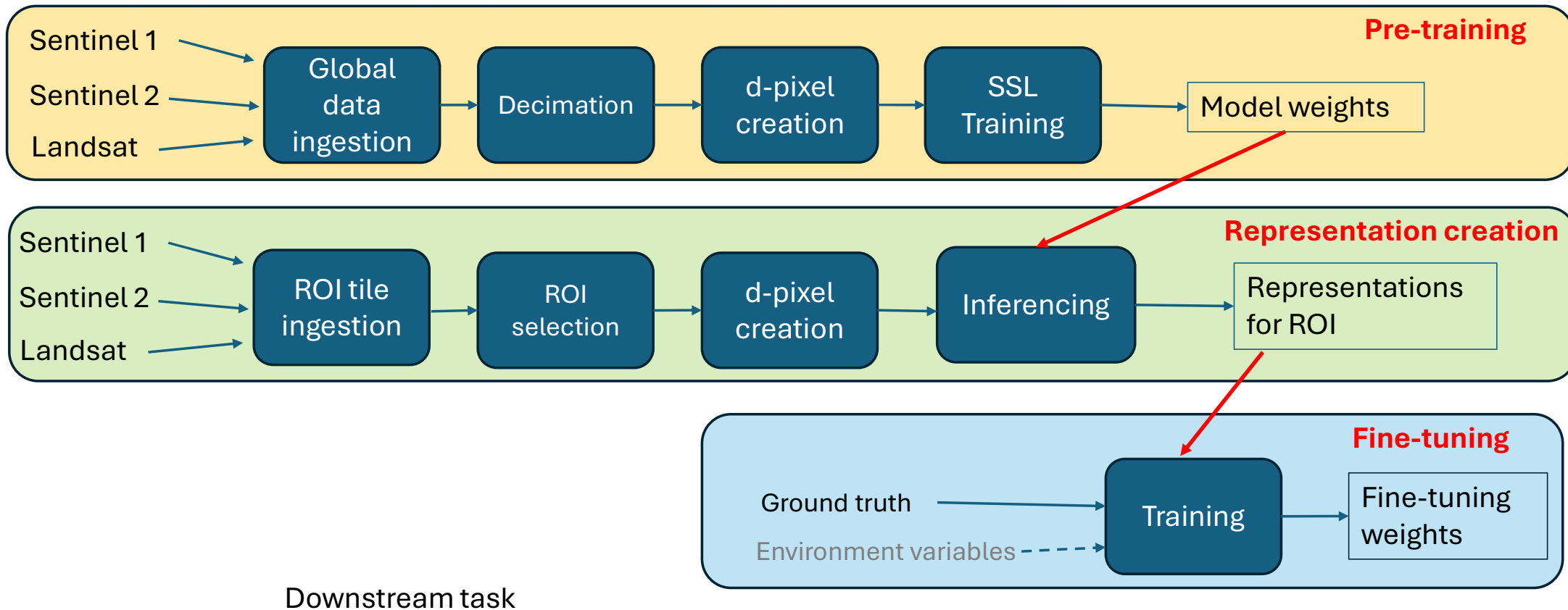
Ground truth

Environment variables

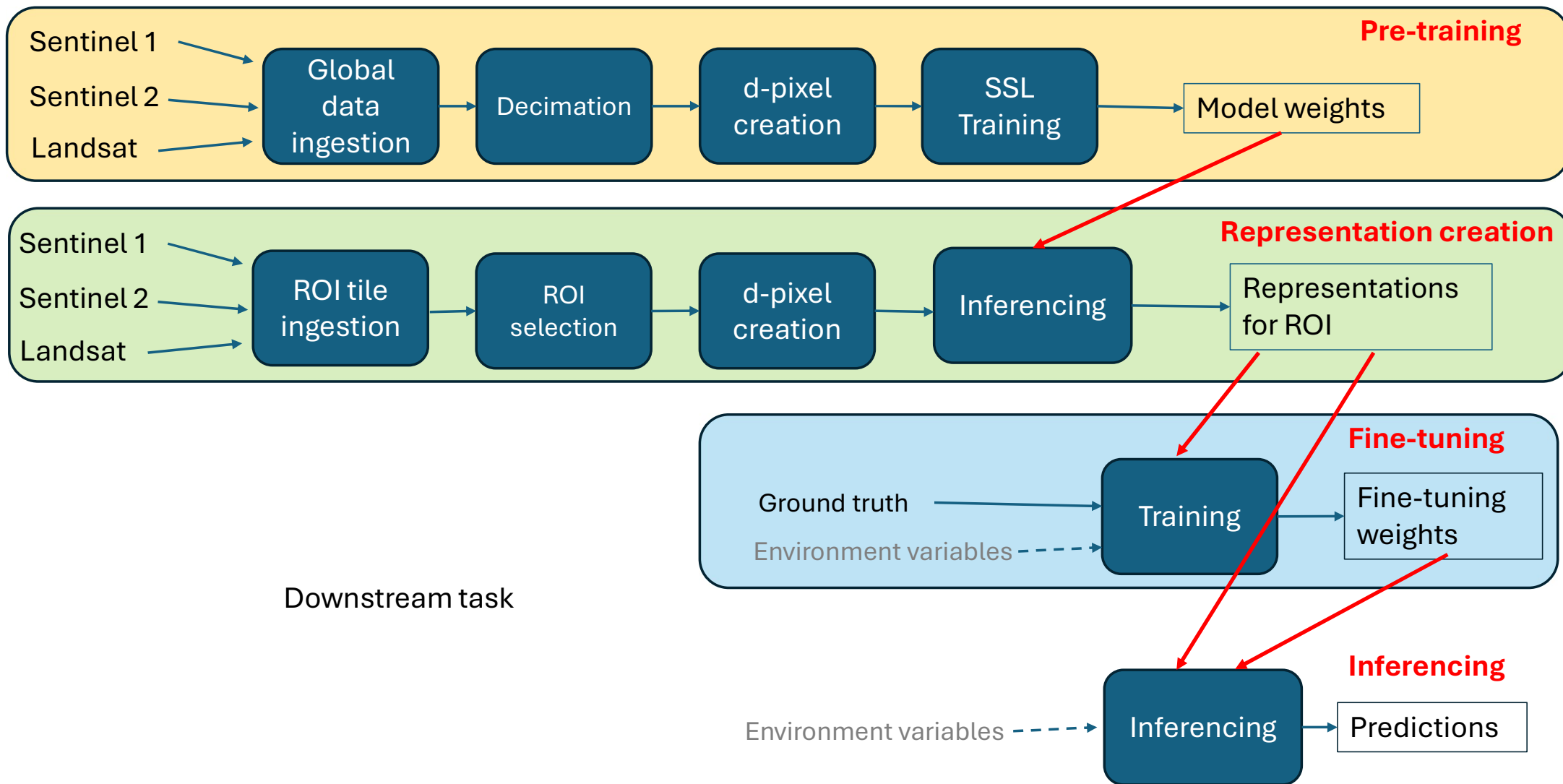
Downstream task

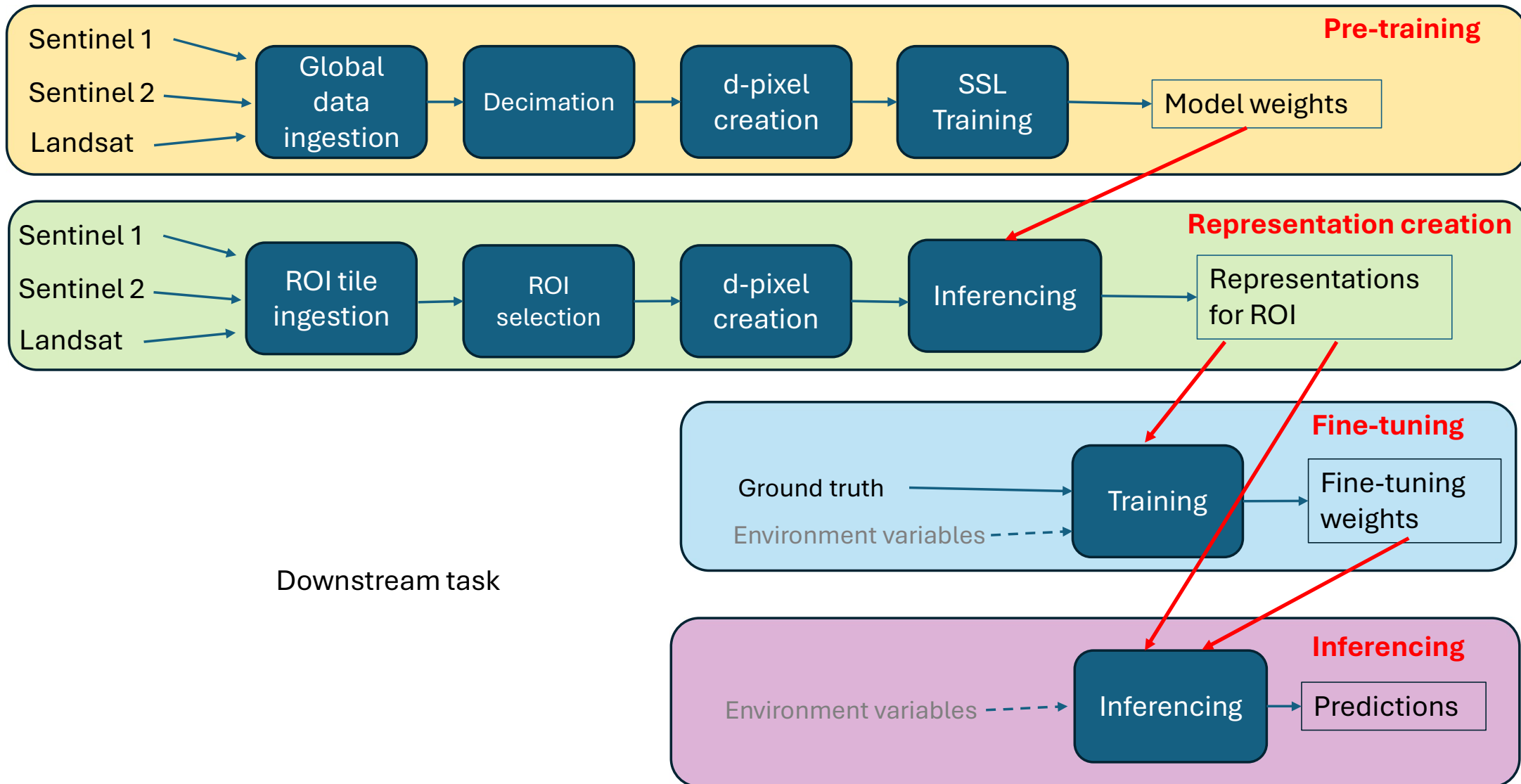


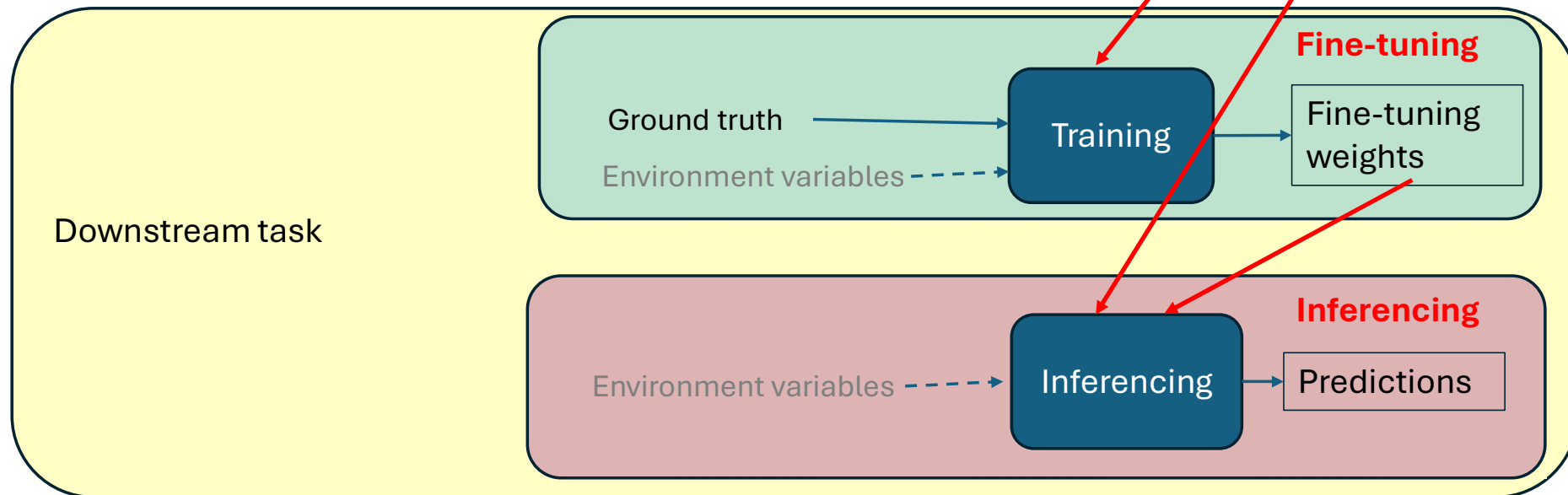
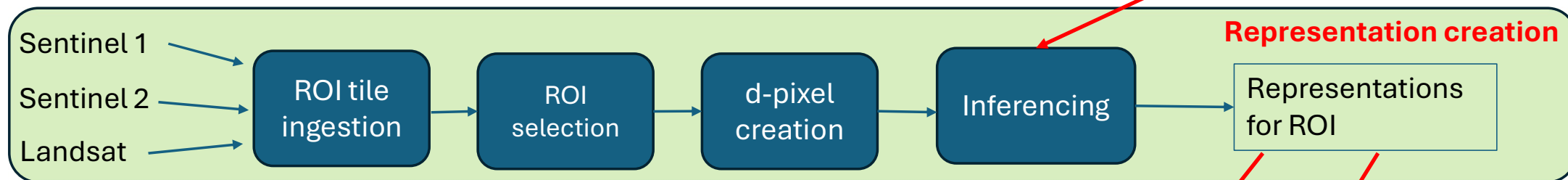
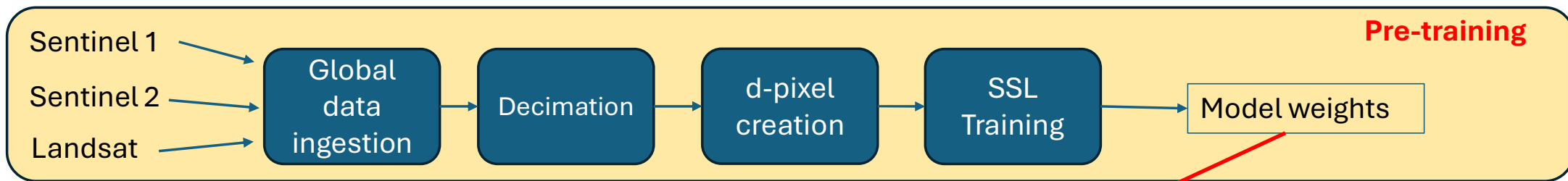


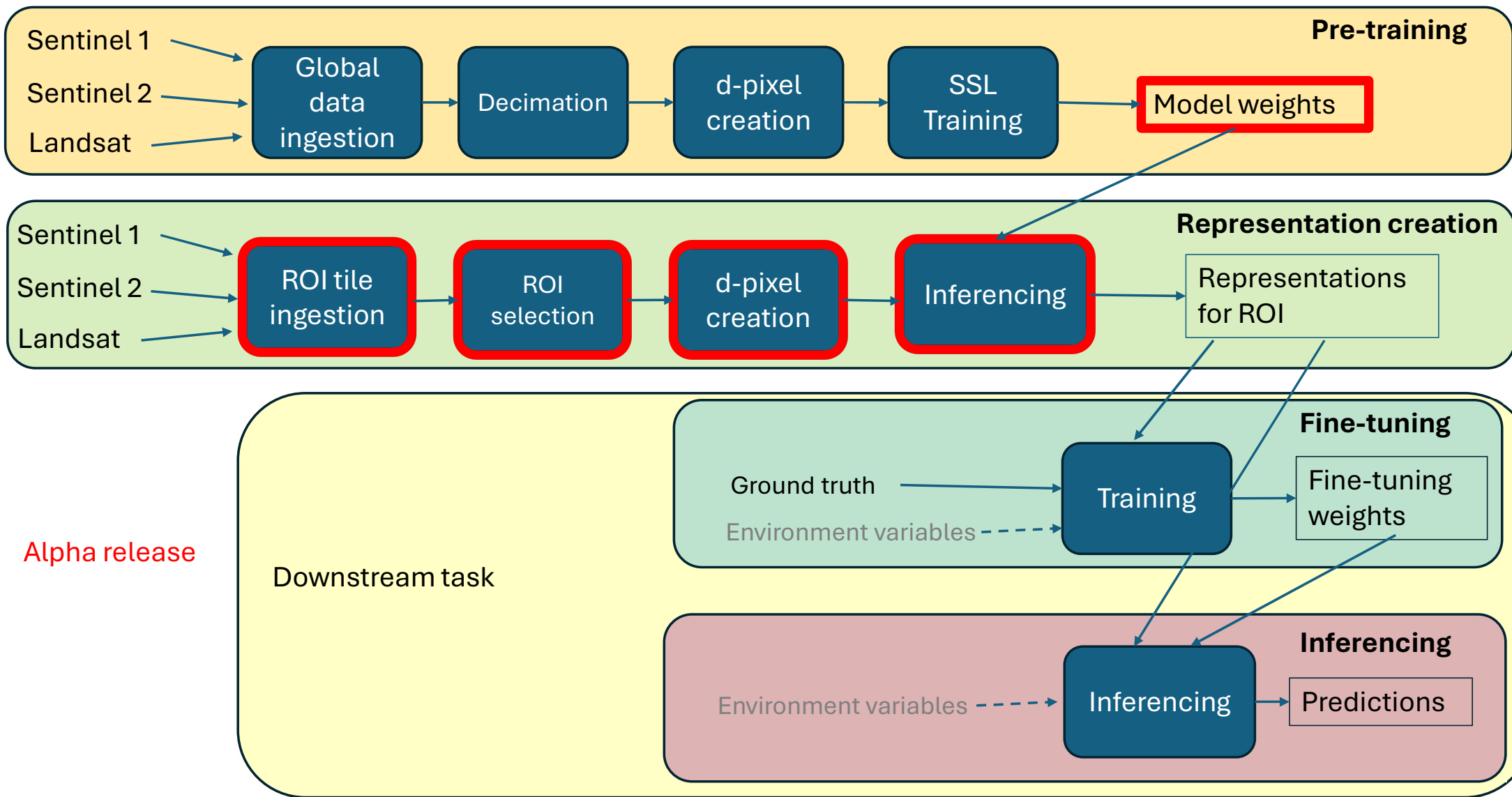


Environment variables $i \rightarrow$

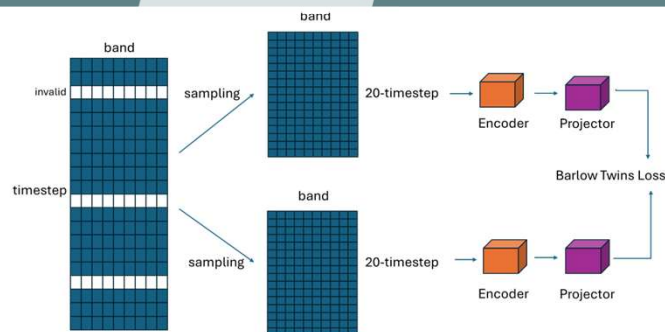








Preliminary Results



Dimension of Representation

Model	Region	Latent Dimension	Out Dimension	Acc	F1
Transformer	Austria	32	512	65.74	63.11
Transformer	Austria	64	512	74.34	71.94
Transformer	Austria	128	512	78.43	77.61
Transformer	Austria	256	512	80.78	79.85

The larger the better!

Dimension of Projected Feature

Model	Region	Latent Dimension	Out Dimension	Acc	F1
Transformer	Austria	128	128	68.15	66.87
Transformer	Austria	128	256	77.58	76.19
Transformer	Austria	128	512	78.43	77.61
Transformer	Austria	128	1024	78.79	78.00

